2025 STAGE Seminar: Weil Conjecture

Yutong Chen

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1 Statement of Weil Conjecture and Proof for Curves

First we give the statement of Weil conjecture.

Theorem 1.1 (Weil Conjecture)**.** *Let X be a scheme of finite type over $\mathbb{F}_q$.*

1. *There are algebraic integers $\alpha_1, \cdots, \alpha_r, \beta_1, \cdots, \beta_s$ such that*

$$\#X(\mathbb{F}_{q^n}) = \alpha_1^n + \cdots + \alpha_r^n - (\beta_1^n + \cdots \beta_s^n)$$

2. *If in addition we assume X is smooth and proper of dimension d, then:*

$$\#X(\mathbb{F}_{q^n}) = \sum_{i=0}^{2d} (-1)^i \sum_{j=1}^{b_i} \alpha_{i,j}^n$$

where

- *$b_i = b_{2d-i}$ are ℓ-adic Betti numbers*
- *$\alpha_{i,j}$ are algebraic integers satsifying $\alpha_{i,j} = q^d/\alpha_{2d-i,j}$*
- *The Riemann Hypothesis: for any archimedean absolute value $|\cdot|$ on $\mathbb{Q}(\alpha_{i,j})$, we have $|\alpha_{i,j}| = q^{\frac{i}{2}}$*

3. *Let R be a finitely generated subring of $\mathbb{C}$. Let $\mathfrak{m}$ be any maximal ideal in R, we know $R/\mathfrak{m}$ is a finite field. Now if X is a smooth proper variety over R, then the Betti numbers b_i in $X_{R/\mathfrak{m}}$ are precisely* $\mathrm{rk}H^i(X(\mathbb{C}), \mathbb{Z})$.

We recall the classical Riemann zeta function $\zeta(s)$ has the following properties:

1. It is holomorphic on $\mathbb{C}$, except for a simple pole at $s = 1$
2. $\zeta(s) = 2^s \pi^{s-1} \sin(\frac{\pi s}{2}) \Gamma(1-s) \zeta(1-s)$
3. I thas zeroes at $-2, -4, \cdots,$; all other zeroes are at $0 \leq \Re s \leq 1$.

Note we can rewrite the Riemann zeta function in the following way:

$$\zeta(s) = \prod (1 - p^{-s})^{-1} = \prod_{\mathfrak{p} \in \text{Spec } \mathbb{Z} \text{ closed points}} (1 - (\#k(\mathfrak{p}))^{-s})^{-1}$$

This motivates the definition of the Riemann zeta function over a variety X:

$$\zeta_X(s) = \prod_{\mathfrak{p} \in X \text{ closed points}} (1 - (\#k(\mathfrak{p}))^{-s})^{-1}$$

Definition 1.2. Let X be a scheme of finite type over $\mathbb{F}_q$. We define

$$Z_X(T) = \exp(\sum \#X(\mathbb{F}_{q^n}) \frac{T^n}{n}) \in \mathbb{Q}[[T]]$$

Remark. *This definition makes sense since $Z_X(q^{-s}) = \zeta_X(s)$*

Hence we can reformulat Weil conjecture in terms of $Z_X(T)$:

1. $Z_X(T) = \frac{(1-\beta_1 T) \cdots (1-\beta_s T)}{(1-\alpha_1 T) \cdots (1-\alpha_r T)}$
2. If X is smooth with dimension d, we can arrange:

$$Z_X(T) = \frac{P_1(T) \cdots P_{2d-1}(T)}{P_0(T) \cdots P_{2d}(T)}$$

   where each $P_i(T) = \prod_{j=1}^{b_i} (1 - \alpha_{i,j} T) \in 1 + T\mathbb{Z}[T]$ with $|\alpha_{i,j}| = q^{\frac{i}{2}}$. Moreover, we have functional equation for $Z_X(T)$:

$$Z_X(\frac{1}{q^d T}) = \pm q^{d\chi/2} T^\chi Z_X(T)$$

   where $\chi = b_0 - b_1 + b_2 - \cdots + b_{2d}$ is the Euler characteristic of X.

3. This remains the same.

We now specialize to the case for curves.

Theorem 1.3 (Weil Conjecture for Curves)**.** *Let X be a smooth, projective, geometrically irreducible over $\mathbb{F}_q$.*

1. $Z_X(T)$ is a rational function

2. There exists $f \in 1 + T\mathbb{Z}[T]$ such that $Z_X(T) = \frac{f(T)}{(1-T)(1-qT)}$, where

$$f(T) = \prod_{j=1}^{2g}(1 - \alpha_j T), |\alpha_j| = \sqrt{q}$$

3. functional equation $Z_X(\frac{1}{qT}) = (qT^2)^{1-g} Z_X(T)$

Proof. (of rationality) We know

$$Z_X(T) = \prod_{\text{closed } P \in X} (1 - T^{\deg P})^{-1} = \sum_{D \geq 0} T^{\deg D} = \sum_{0 \leq \deg D \leq 2g-2} T^{\deg D} + \sum_{\deg D \geq 2g-2} T^{\deg D}$$

Now for $d \geq 2g-1$, and D is an effective divisor of degree d. Then the number of effective divisors linearly equivalent to D is the size of the complete linear system $|D| \simeq \mathbb{P}(\Gamma(X, \mathcal{L}(D)))$, which is $\frac{q^{l(D)}-1}{q-1}$. By Riemann Roch: $l(D) = d - g + 1 \implies |D|$ has cardinality $\frac{q^{d-g+1}}{q-1}$. We also know in this case every divisor of degree d is linearly equivalent to an effective one. So we have:

$$\sum_{deg D \geq 2g-2} T^{\deg D} = |\mathrm{Cl}^0(X)| \sum_{d=2g-2}^{\infty} \frac{q^{d-g+1} - 1}{q-1} T^d$$

which simplifies to

$$\frac{|\mathrm{Cl}^0(X)|}{q-1} \cdot \frac{(q^g - 1)T^{2g-1} + (q - q^g)T^{2g}}{(1-T)(1-qT)}$$

is rational. □

Combining with the first part, we know we can write:

$$Z_X(T) = \frac{\prod_{i=1}^{2g}(1 - \alpha_i T)}{(1-T)(1-qT)}$$

with $\prod \alpha_i = q^g$.

The second part follows from the so called Hasse-Weil bound:

Theorem 1.4. *Let $N = \#C(\mathbb{F}_q)$, where C is a geometrically irreducible smooth projective curve of genus g. Then:*

$$|N - (q+1)| \leq 2g\sqrt{q}$$

Proof. (Hasse-Weil implies Riemann Hypothesis for Curves) Write $N_r = \#C(\mathbb{F}_{q^r})$, then

$$|N_r - (q^r + 1)| \leq 2g\sqrt{q^r}$$

We wish to show $|\alpha_i| = \sqrt{q}$. Since we know $\prod \alpha_i = q^g$, it suffices to show $|\alpha_i| \leq \sqrt{q}$. Note by definition:

$$\sum_{r \geq 1} N_r t^{r-1} = \frac{d}{dt} \log Z_X(t) = \sum_{i=1}^{2g} -\frac{\alpha_i}{1 - \alpha_i t} + \frac{1}{1-t} + \frac{q}{1-qt}$$

$$= \sum_{r \geq 1} (q^r + 1 - \sum_{i=1}^{2g} \alpha_i^r) t^{r-1}$$

By Hasse-Weil, we have

$$|\sum \alpha_i^r| \leq 2g\sqrt{q^r}$$

Now set $a_r = \sum \alpha_i^r$. We have $\sum a_r t^r = \sum_{i=0}^{2g} \frac{\alpha_i t}{1-\alpha_i t}$. We may assume $|\alpha_1| > \sqrt{q}$ maximal. We take $t \to \frac{1}{|\alpha_i|}$, then LHS converges but RHS diverges. Contradiction. □

The remaining goal is to prove Hasse-Weil. The main tool is intersection theory.

Theorem 1.5. *Let X be a smooth projective surface over $k = \overline{k}$. Then there exists a unique symmetric bilinear pairing (called intersection product)*

$$\mathrm{Div}(X) \times \mathrm{Div}(X) \to \mathbb{Z}$$

such that if two curves intersecting transversely, $(C, C') = \#(C \cap C')$.

Remark. *Intersection number is invariant up to linear equivalence.*

There are several important properties of intersection product.

Proposition 1.6. *Let C be a curve in a surface X. Then:*

$$C^2 = \deg_C N_{C/X}$$

Remark. *In particular, if C is a genus g curve. And let $\Delta : C \to C \times C$ is diagonal map. Then $\Delta \cdot \Delta = \deg_C(\omega_C^\vee) = -(2g-2) = 2-2g$*

The following is a direct consequence of Serre duality:

Proposition 1.7 (Adjunction Formula)**.** *Let C be a genus g curve inside surface X. Then:*

$$2g - 2 = C \cdot (C + K_X)$$

Let Γ be the graph of the Frobenius map. We have a commutative diagram:

$$\begin{array}{ccc} C & \longrightarrow & C \\ \downarrow{\scriptstyle f^*\Delta} & & \downarrow{\scriptstyle \Delta} \\ C \times C & \xrightarrow[f=\mathrm{id}\times\mathrm{Frob}]{} & C \times C \end{array}$$

So we can compute:

$$\Gamma^2 = (f^*\Delta, f^*\Delta) = (f_* f^*\Delta, \Delta) = \deg(f)(\Delta, \Delta) = q(2-2g)$$

Moreover, since Δ and Γ intersect transversely, $\Delta \cdot \Gamma = N$.

Finally we need the following corollary of Hodge Index Theorem:

Theorem 1.8. *Let $X = C \times C'$, for C, C' smooth projective curves. Consider the divisor*

$$l = C \times \{*\}, m = \{*\} \times C'$$

If $D \in \mathrm{Div}(X)$ such that $D \cdot l = a$, $D \cdot m = b$. Then $D^2 \leq 2ab$.

Proof. (of Hasse-Weil bound) Apply to $D = r\Gamma + s\Delta$. We compute $D \cdot l = rq + s, D \cdot m = r + s$. Therefore we have

$$(r^2q + s^2)(2 - 2g) + 2rsN = D^2 \leq 2(rq + s)(r + s)$$

Rearranging, gives:

$$|N - q - 1| \leq |\frac{rq}{s} + \frac{sg}{r}|$$

We then take r, s accordingly. □

2 Étale Site

2.1 definition

Definition 2.1. A morphism $f : X \to S$ is étale if it's flat, unramified, and locally of finite presentation.

The first goal is to explain the adjective "unramified".

Let B/A be an extension of DVRs, with uniformizers π_B and π_A. We say B/A is unramified if π_A generates (π_B) as an ideal, and $B/(\pi_B)$ over $A/(\pi_A)$ is a finite separable extension. This description has an obvious generalization in the context of local rings:

Definition 2.2. A local homomorphism of local rings $f : A \to B$ is unramified if $f(\mathfrak{m}_A)B = \mathfrak{m}_B$ and $B/\mathfrak{m}_B$ is a finite separable extension over $A/\mathfrak{m}_A$

Definition 2.3. A morphism $f : X \to S$ is unramified at $x \in X$ if f is locally of finite type at x and $\mathcal{O}_{S,f(x)} \to \mathcal{O}_{X,x}$ is an unramified homomorphism of local rings. We say f is unramified if it's unramified at all points.

Example. *Open and closed immersions are unramified.*

Example. *Consider $f : \mathbb{A}^1 \to \mathbb{A}^1$ given by $t \mapsto t^2$. This corresponds to ring map $k[x] \hookrightarrow k[\sqrt{x}]$. Then f ramifies at $0 \in \mathbb{A}^1$. Indeed, the local ring morphism is*

$$k[x]_{(x)} \to k[\sqrt{x}]_{\sqrt{x}}$$

but (x) does not generate $(\sqrt{x})$.

There is an equivalent definition for étale morphisms:

Proposition 2.4. *A morphism $f : X \to S$ is étale at a point $x \in X$ if it is smooth of relative dimension 0 at x.*

Again to make sense of this, we recall the definition(s) of smoothness.

The most explicit characterization of smoothness is the following:

Definition 2.5. Let $r \in \mathbb{N}$, $f : X \to S$ be a morphism of schemes and let $x \in X$. We say f is smooth of relative dimension r at x if and only if in a Zariski neighborhood of x it's locally given by form:

$$\text{Spec } \frac{A[t_1, \cdots, t_n]}{(g_{r+1}, \cdots, g_n)} \to \text{Spec } A$$

such that the matrix $(\frac{\partial g_i}{\partial t_j})(x)$ has full rank. We say f is smooth if it's smooth of some relative dimension at all $x \in X$.

Here is an equivalent definition from EGA:

Definition 2.6. A morphism of schemes $f : X \to S$ is smooth if:

1. f is flat
2. f is locally of finite presentation,
3. for all $s \in S$, the fibres X_s over $k(s)$ is geometrically regular.

For $x \in X$, the morphism f is called smooth at x is there is an open neighborhood U of x such that $f|_U$ is smooth.

We would like to make use of the sheaf of differentials when smooth morphisms are presented:

Proposition 2.7. *Let $f : X \to S$ be a flat morphism between irreducible k-varieties. Let $r = \dim X = \dim S$. Then f is smooth of relative dimension r if and only if $\Omega_{X/S}$ is locally free of rank r.*

2.2 Properties and Examples

First note the property of being étale is stable under base change.
The prototypical étale morphism is the following:

Definition 2.8. Let A be a ring and $p \in A[t]$ be a monic polynomial. Let $B = A[t]/(p)$and $C = B[q^{-1}]$ for some $q \in B$. Now suppose $p'(t) \in C$ is in $C^\times$, then Spec $C \to$ Spec A is an étale morphism. This is called the standard étale morphism.

Example. *Let $A = k[x], B = k[x, y]/(y^2 - (x^3 - x)), C = B[y^{-1}] = k[x, y, y^{-1}]/((y^2 - (x^3 - x)))$, then $p'(y) = 2y \in C^\times$. So the projection* Spec $C \to$ Spec A *is étale. Geometrically, this is the projection from the elliptic curve $E : y^2 = x^3 - x$ missing three points on the x-axis to the affine line.*

In fact standard étale morphism can be taken as the local model for étale morphisms.

Proposition 2.9 (Local structure of an étale morphism)**.** *Let $f : X \to S$ be a morphism of schemes. Let $x \in X$ and $s = f(x) \in S$. Then f is étale at x if and only if it's locally given by a standard étale morphism.*

We also formally introduce the notion of local coordinates.

Proposition 2.10. *Let $f : X \to S$ be smooth of relative dimension r at a point $x \in X$, $s = f(x)$. Then the following are equivalent for elements $g_1, \cdots, g_r \in \mathcal{O}_{X,x}$:*

1. *the differentials $dg_1(x), \cdots dg_r(x)$ form a basis for the cotangent space $\Omega_{X/S,x} \otimes k(x)$*
2. *There is an open neighborhood U of x in X to which the g_i etend, such that the S-morphism*
$$(g_1, \cdots, g_r) : U \to \mathbb{A}^r_S$$
is étale.

Moreover, such $g_1, \cdots, g_r$ exists. Such r-tuple is called a system of local coordinates at x.

Remark. *This says any smooth map $X \to S$ locally factorizes as*
$$X \to \mathbb{A}^r_S \to S$$
where the first morphism is étale. This enables us to prove the fact that the property of reducedness, normality, regularity are all preserved under smooth maps.

2.3 Étale Fundamental Group

Fix a nice topological space X. We have an equivalence of categories:
$$F_x : \mathsf{Cover}_X \to \mathsf{Set}, \quad (p : X \to Y) \mapsto p^{-1}(x)$$

Theorem 2.11. $\pi_1(X, x) \simeq \mathrm{Aut}(F_x)$

This shows that we can actually define the fundamental group in terms of the automorphism of the fibre functor.

There's a natural variant of the above situation which suits the algebraic picture better: we replace the categories of covers by finite covers. So we have a functor:
$$F_x|_{\mathsf{FCover}_X} \colon \mathsf{FCover}_X \to \mathsf{Set}$$
and the "fundamental group" is this setting should be $\mathrm{Aut}(F_x|_{\mathsf{FCover}_X})$, whish turns out to be the pofinite completion of the fundmamental group $\widehat{\pi}_1(X, x)$: the inverse limit of all finite quotients of $\pi_1(X, x)$

We now want to define the fundamental group in the algebraic setting. Let's consider a basic example $X = \mathbb{C}^\times$. Topologically $X \simeq S^1$ so "should" have fundamental group $\mathbb{Z}$. However, note that the infinite cover $\exp : \mathbb{C} \to \mathbb{C}^\times$ is non-algebraic, this hints us that we may only restrict ourselves to the finite covers, hence the answer should be $\widehat{\mathbb{Z}}$.

Now the analogue of covering in the context of algebraic geometry is an étale map, we thus naturally have the following definition.

Definition 2.12. Let $x : \mathrm{Spec}\ \Omega \to X$ be a geometric point of X. Given a finite étale morphism $p : Y \to X$, we can pull back along x to get a finite disjoing unions of copies of Spec (Ω), and forgetting the scheme structure gives us a set $p^{-1}(x)$. The fibre functor is:
$$F_x : \mathsf{FEt}_X \to \mathsf{Set}, \quad (p : Y \to X) \mapsto p^{-1}(x)$$

The *étale fundamental group* is $\pi_1^{\mathrm{et}}(X, x) = \mathrm{Aut}(F_x)$. This also gives us a notion of simply connectedness in this case.

This definition is justified by the following theorem:

Theorem 2.13 (Generalized Riemann Existence Theorem)**.** *For each $\mathbb{C}$-variety X, the natural functors*

$$\mathsf{FEt}_X \to \mathsf{FEt}_{X^{\mathrm{an}}} \to \mathsf{FCover}_{X(\mathbb{C})}$$

are equivalence of categories.

Here is the main theorem for the theory of étale fundamental groups:

Theorem 2.14 (Grothendieck)**.** *Let X be a connected scheme, x a geometric point of X.*

1. *The group $\pi_1(X, x)$ is profinite, and its action on $F_x(Y)$ is continuous for every étale cover $p : Y \to X$*
2. *The functor F_x induces an equivalence of FEt_X with the category of finite continuous $\pi_1(X, x)$-sets. Connected covers correspond to sets with transitive fundamental group action, and Galois covers to finite quotients of $\pi_1(X, x)$.*

Example. *We work out the easiest example for $X = \operatorname{Spec} k$ is a field. A finite étale X-scheme is nothing but the spectrum of a finite étale k-algebra. It suffices to consider only the connected covers, namely* Spec L *for L finite separable extensions of k.*

Choose x a geometric point in X (which amounts to choose an algebraic closure of k). Now the underlying set of Spec $(L \otimes_k \Omega)$ *is a finite set indexed by k-algebra homomorphisms $L \to \Omega$. Note the image must lie in k_s, the separable closure of k in Ω given by the emebdding x. One sees $F_x(X) \simeq \operatorname{Hom}_k(L, k_s)$ for all connected covers* Spec L*, and therefore:*

$$\pi_1(X, x) \simeq \operatorname{Gal}(k_s/k)$$

In particular, the above theorem says $\mathsf{FEt}_{\operatorname{Spec} k}$ is equivalent to the category of continuous $\operatorname{Gal}(k_s/k)$*-sets.*

2.4 Grothendieck Topologies

Definition 2.15. Let $\mathcal{C}$ be a category. Consider all families of morphisms $\{U_i \to U\}_{i \in I}$ in $\mathcal{C}$ having a common target. A *Grothendieck topology* on $\mathcal{C}$ is a set $\mathcal{T}$ consisting families of these (they're called open coverings) such that:

1. Isomorphisms are open coverings
2. If $\{U_i \to U\} \in \mathcal{T}$, $\{V_{ij} \to V_i\} \in \mathcal{T} \implies \{V_{ij} \to U\} \in \mathcal{T}$, "open covering of an open covering is an open covering"
3. $\{U_i \to U\} \in \mathcal{T} \implies \{V \times_U U_i \to V\} \in \mathcal{T}$, "base change of an open covering is an open covering"

A *site* is a pair $(\mathcal{C}, \mathcal{T})$.

Example (Zariski site)**.** *Let X be a topological space, $\mathcal{C} = \mathsf{Open}(X)$, $\mathcal{T} = \{\{U_i \to U\} \mid \bigcup U_i = U\}$. This is the classical Grothendieck topology. For X a scheme, we denote this as X_{Zar} the Zariski site.*

Definition 2.16 (Étale Site)**.** Fix a scheme X, $\mathcal{C} = \mathsf{Et}_X$ be the category of etale spaces over X:

1. objects: schemes U together with étale $U \to X$
2. morphisms: X-morphisms (note these are automatically étale)

A family $\varphi_i : U_i \to U$ is an open covering if $\bigcup \varphi_i(U_i) = U$.

Definition 2.17 (fppf and fpqc sites)**.** Fix a scheme X, $\mathcal{C} = \mathsf{Sch}_X$ be the category of schemes over X. We define $\{\varphi_i : U_i \to U\} \in \mathcal{T} \iff \coprod U_i \to U$ is fppf/fpqc.

Definition 2.18. A continuous map between sites $(\mathcal{C}', \mathcal{T}') \to (\mathcal{C}, \mathcal{T})$ is a functor $F : \mathcal{C} \to \mathcal{C}'$ respecting open covering:

1. $\{U_i \to U\} \in \mathcal{T} \implies \{FU_i \to FU\} \in \mathcal{T}$
2. $\{U_i \to U\} \in \mathcal{T}$ and $V \to U$ in $\mathcal{C}$, then:

$$F(V \times_U U_i) \to FV \times_{FU} FU_i$$

   is an isomorphism.

Remark. *Let $f : X' \to X$ be a continuous map. Then the functor $f^{-1} : \mathsf{Open}(X) \to \mathsf{Open}(X')$ gives a continuous map between the classical Grothendieck topologies. Indeed:*

$$f^{-1}(V \times_U U_i) = f^{-1}V \times_{f^{-1}U} f^{-1}U_i$$

Recall from algebraic geometry that Zariski open covering $\implies$ étale and surjective $\implies$ fppf $\implies$ fpqc. Hence we have continuous maps of sites:

$$X_{\text{fpqc}} \to X_{\text{fppf}} \to X_{\text{ét}} \to X_{\text{Zar}}$$

We now want to develop sheaf theory on sites. Recall for any category $\mathcal{C}$, we can defined a presheaf to be a functor $\mathcal{C}^{\text{op}} \to \mathsf{Ab}$. Now imitating the definition of sheaves on a topological space, we have the following definition:

Definition 2.19. Let $\mathcal{F}$ be a presheaf on a site $(\mathcal{C}, \mathcal{T})$. Then we call $\mathcal{F}$ is a sheaf if

$$\mathcal{F}(U) \to \prod_i \mathcal{F}(U_i) \rightrightarrows \prod_{i,j} \mathcal{F}(U_i \times_U U_j)$$

is exact for all open coverings $\{U_i \to U\}$ in $\mathcal{T}$.

Remark. *If our site is $X_{et}, X_{fpqc}, X_{fppf}$, and suppose we have an open covering $\{U_i \to U\}$. We may consider $U' = \coprod U_i$, $U'' = U' \times_U U'$ and the sheaf axiom becomes*

$$\mathcal{F}(U) \to \mathcal{F}(U') \rightrightarrows \mathcal{F}(U'')$$

Hence we only need to check for open covering of form $\{U' \to U\}$ consisting of a single morphism.

Now we give an important construction. Let X be a scheme, $\mathcal{C}$ be a subcategory of Sch_X, $\mathcal{F} \in \mathsf{QCoh}(X)$. We can define a presheaf (of abelian groups) on $\mathcal{C}$ as follows: for each object $U \xrightarrow{p} X$ in $\mathcal{C}$, we define $\mathcal{F}_{\mathcal{C}}(U) = (p^*\mathcal{F})(U) = \mathrm{Hom}(p^*\mathcal{O}_X, p^*\mathcal{F})$.

Proposition 2.20. *In fact, the above construction gives a sheaf.*

To prove this, we have a brief detour to an important property of fpqc morphisms. Suppose $S = \cup S_i$, we set $S' = \coprod S_i$ and let $\pi : S' \to S$ be the natural map. A natural question to ask when does a sheaf on S' is the pullback of a sheaf on S. The answered is simply the gluing conditions. Let $S'' = S' \times_S S'$, p_1, p_2 be two projections. Then we need to have an isomorphism $\varphi : p_1^{-1}\mathcal{F} \to p_2^{-1}\mathcal{F}$ satisfying the cocycle condition, which reads: if $S''' = S' \times_S S' \times_S S'$, then $p_{13}^{-1}\varphi = p_{23}^{-1}\varphi \circ p_{12}^{-1}\varphi$. This motivates the following definition:

Definition 2.21 (Descent Data)**.** Let $p : S' \to S$ be fpqc morphism, and $\mathcal{F}'$ is a quasi-coherent sheaf on S'. A descent datum on F' is an isomorphism $\varphi : p_1^*\mathcal{F}' \to p_2^*\mathcal{F}'$ of S''-modules satisfying the cocylce condition: $p_{13}^*\varphi = p_{23}^*\varphi \circ p_{12}^*\varphi$. A morphism of descent data $(\mathcal{F}', \varphi) \to (\mathcal{G}', \psi)$ is morphism $f : \mathcal{F}' \to \mathcal{G}'$ making the following diagram commute:

$$\begin{array}{ccc} p_1^*\mathcal{F}' & \xrightarrow{p_1^*f} & p_1^*\mathcal{G}' \\ {\scriptstyle\varphi}\downarrow & & \downarrow{\scriptstyle\psi} \\ p_2^*\mathcal{F}' & \xrightarrow[p_2^*f]{} & p_2^*\mathcal{G}' \end{array}$$

Note that if $\mathcal{F}$ is a S-module, then $p^*\mathcal{F}$ has a natural descent datum. In fact, the converse is also true:

Theorem 2.22 (Grothendieck)**.**

$$p^* : \mathsf{QCoh}(S) \to \{\mathsf{QCoh}(S') \text{ with descent data}\}$$

is an equivalence of categories.

Now we can prove our original proposition:

Proof. Let $p : S' \to S$ be a fpqc morphism of X-schemes. Set $S'' = S' \times_S S'$, we need to check the exactness of

$$\mathcal{F}_{\mathcal{C}}(S) \to \mathcal{F}_{\mathcal{C}}(S') \rightrightarrows \mathcal{F}_{\mathcal{C}}(S'')$$

If we set $q : S'' \to S$ be the natrual map, and rename $\mathcal{F}$ as its pullback to S, then it's equivalent to check the exactness of

$$\mathrm{Hom}(\mathcal{O}_S, \mathcal{F}) \to \mathrm{Hom}(p^*\mathcal{O}_S, p^*\mathcal{F}) \rightrightarrows \mathrm{Hom}(q^*\mathcal{O}_S, q^*\mathcal{F})$$

But this is just part of Grothendieck's theorem. Note the same proof works for étale or fppf sites. □

Proposition 2.23. *Let S be a scheme, and let X, Y be S-schemes. The functor $U \mapsto \mathrm{Hom}_U(X_U, Y_U)$ is a sheaf of sets on S_{fpqc}, denoted as $\mathbf{Hom}(X, Y)$. In particular, if we consider $X = Y$, then the functor $U \mapsto \mathrm{Aut}_U(X_U)$ is a sheaf of groups.*

Proof. Similar as above. □

Remark. *Another example of sheaf (of sets) on S_{fppf} is the functor of point. Let X be a S-scheme, then $U \mapsto X(U)$ is a sheaf.*

Proposition 2.24. *Let $\mathcal{F}$ be a presheaf on a site. Then the sheafification of $\mathcal{F}$ exists.*

We can thus accordingly define sheaf kernel and sheaf images.

3 Étale Cohomology

Let X is a qcqs scheme.

3.1 Sheaf Theory

Definition 3.1. An étale sheaf is a presheaf $\mathcal{F}$ on the small étale site $X_{\text{ét}}$ satisfying the sheaf axiom.

Example. *(Constan Sheaf) If A is an abelian gorup, we can define the constan sheaf $\underline{A}$ sends an étale open $U \to X$ to $\mathrm{Hom}(U, A)$ where we endow A the discrete topology.*

Example. *Given $\varphi : \mathcal{F}' \to \mathcal{F}$ of étale sheaves, we can define the sheaf kernel.*

There are two ways to build sheaves: sheafification and descent.

Theorem 3.2 (Étale Descent)**.** *There is a functor $\mathsf{QCoh}(X) \to \mathrm{PSh}(X_{\text{ét}})$,*

$$(\mathcal{F} \in \mathsf{QCoh}(X)) \mapsto (p : U \to X) \mapsto \mathcal{F}_{\text{ét}}(U) = \Gamma(U, p^*\mathcal{F})$$

In fact:

1. *$\mathcal{F}_{\text{ét}} \in \mathrm{Shv}(X_{\text{ét}})$*
2. *This is fully faithfully*
3. *Its essential image is quasicoherent sheaves on $X_{\text{ét}}$*

Example. *Suppose I have a morphism of schemes $Y \to X$, the functor of point $U \mapsto Y(U)$ is an étale sheaf (of sets). As a concrete example, $\mathbb{G}_{m,X}(U) = \mathcal{O}_U(U)^\times$ is an étale sheaf.*

Given a geometric point $\overline{x} \to X$. We can define the stalk

$$\mathcal{F}_{\overline{x}} = \operatorname*{colim}_{\overline{x} \hookrightarrow U, U \text{ étale over } X} \mathcal{F}(U)$$

We can check exactness on stalks.

3.2 Cohomology

Definition 3.3.

$$H^i(X_{\text{ét}}, -) = R^i\Gamma(X_{\text{ét}}, -)$$

Example. *Let $X/\overline{k}$.*

$$H^0(X_{\text{ét}}; F) = \Gamma(X_{\text{ét}}; F)$$

Definition 3.4. Given $f : X \to Y$ a morphism of schemes. We have pushforward of étale sheaves:

$$f_* : \text{Shv}(X_{\text{ét}}) \to \text{Shv}(Y_{\text{ét}}), \quad f_*F(U) = F(U \times_Y X)$$

Remark. *pushforward is left exact. Hence we can define higher pushforward.*

Example. *Let $X = \text{Spec } k$ be a point. We have an equivalence of categories*

$$\text{Shv}(X_{\text{ét}}) \simeq \text{Mod-}(\text{Gal}(k_s/k))$$

sending F to $F(k_s)$. In fact

$$H^i(X_{\text{ét}}, F) = H^i(\text{Gal}(k_s/k), F(k_s))$$

Proposition 3.5.

$$H^1(X_{\text{ét}}, \mathbb{G}_m) = \text{Pic}(X)$$

Theorem 3.6. *Let $k = \overline{k}$, X/k be a smooth projective irreducible curve.*

$$H^i(X_{\text{ét}}, \mathbb{G}_m) = \begin{cases} k^\times, & \text{if } i = 0; \\ \text{Pic}(X), & \text{if } i = 1; \\ 0, & \text{if } i = 2; \\ 0, & \text{otherwise.} \end{cases}$$

Corollary 3.7. *Let* $\text{char } k = 0$.

$$H^i(X_{\text{ét}}, \mu_n) = \begin{cases} \mu_n, & \text{if } i = 0; \\ \text{Pic}^0(X)[n], & \text{if } i = 1; \\ \mathbb{Z}/n\mathbb{Z}, & \text{if } i = 2; \\ 0, & \text{otherwise.} \end{cases}$$

Definition 3.8. Fix a smooth projective variety over a field k. For each prime ℓ coprime to the characteristic of k, we can define the ℓ-adic cohomology of X:

$$H^i_\ell(X, \mathbb{Q}_\ell) = \lim H^i((X_{\overline{k}})_{\text{ét}}, \underline{\mathbb{Z}/\ell^\bullet\mathbb{Z}}) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$$

Remark. *Suppose X has genus g. Then recall:*

$$H^i(X(\mathbb{C}), \mathbb{Z}) = \begin{cases} \mathbb{Z} & \text{if } i = 0; \\ \mathbb{Z}^{2g}, & \text{if } i = 1; \\ \mathbb{Z}, & \text{if } i = 2; \\ 0, & \text{otherwise.} \end{cases}$$

Note then our ℓ-adic cohomology spaces gives the right $\mathbb{Q}_\ell$-dimension.

3.3 Weil Cohomology Theory

Let k be a field (not necessarily algebraically closed). We denote $P(k)$ the be the class of smooth projective varieties over k. Let's fix another field F with $\mathrm{char}(F) = 0$

Definition 3.9 (Weil Cohomology Data)**.** A Weil cohomology theory consists of the following data:

- $F(1)$ a one-dimensional vector space over F
- A contravariant functor $H^\bullet : P(k)^{\mathrm{op}} \to \mathrm{GrAlg}_F$
- For X of equidimension d, we need a trace map:

$$\int_X : H^{2d}(X)(d) \to F$$

  where $H^{2d}(X)(d) = H^{2d}(X) \otimes F(1)^{\otimes d}$

- For each $X \in P(k)$, a cycle class map

$$\mathrm{cl}_X : \mathrm{CH}^i(X) \to H^{2i}(X)(i)$$

  which is required to be a group homomorphism.

Remark. *Suppose I have $f : X \to Y$, then we denote $f^* : H^*(Y) \to H^*(X)$. This is required to be a ring homomorphism, so*

$$f^*(\alpha \cup \beta) = f^*(\alpha) \cup f^*(\beta)$$

To form a Weil Cohomology theory, we need to add a lot of compatibility conditions.

Definition 3.10 (Künneth Formula)**.** Fix a Weil cohomology datum. We say it satisfies Künneth formula if

1. We have the Künneth isomorphism:

$$H^\bullet(X) \otimes H^\bullet(Y) \to H^\bullet(X \times Y)$$

$$\alpha \otimes \beta \mapsto \mathrm{pr}_1^* \alpha \cup \mathrm{pr}_1^* \beta := \alpha \boxtimes \beta$$

2. Fubini Theorem: if $\alpha \in H^{2d}(X)(d), \beta \in H^{2e}(Y)(e)$ for $d = \dim X, e = \dim Y$, then

$$\int_{X \times Y} \alpha \boxtimes \beta = \int_X \alpha \cdot \int_Y \beta$$

Definition 3.11 (Poincaré Duality)**.** We say a Weil cohomology datum satisfies Poincaré duality if:

1. If $\dim X = d$, then the pairing

$$H^i(X) \times H^{2d-i}(X)(d) \xrightarrow{\cup} H^{2d}(X)(d) \xrightarrow{\int_X} F$$

   is perfect.

2. $\dim H^i(X) < \infty$, for each i.

With Poincaré duality, we can define the pushforward:

Definition 3.12. If X and Y have dimensions d and e. We define

$$f_* : H^{2d-i}(X)(d) \to H^{2e-i}(Y)(e)$$

as the adjoint of f^*. More formally,

$$\int_X f^*\beta \cup \alpha = \int_Y \beta \cup f_*\alpha$$

Lemma 3.13 (Projection Formula)**.**

$$f_*(f^*\beta \cup \alpha) = \beta \cup f_*\alpha$$

Definition 3.14 (Cycle Coherence)**.** Fix a Weil cohomology datum satisfying Poincaré duality. We say it satisfies cycle coherence if:

1. It satisfies pushforward formula: if $f : X \to Y$, $\mathrm{cl}_Y(f_*\alpha) = f_*(\mathrm{cl}_X(\alpha))$
2. It satisfies pullback formula: $\mathrm{cl}_X f^!\beta = f^*\mathrm{cl}_Y\beta$
3. $\mathrm{cl}_X(\alpha \cdot \alpha') = \mathrm{cl}_X\alpha \cup \mathrm{cl}_X\alpha'$
4. Nondegenaracy:

$$\int_{\mathrm{Spec}\ k} \mathrm{cl}_{\mathrm{Spec}\ k}(\mathrm{Spec}\ k) = 1$$

Definition 3.15. A Weil cohomology datum satisfying Künneth formula, Poincaré duality, and cycle coherence is called a *pre-Weil Cohomology Theory*

Example. *Fix a pre-Weil cohomology theory. If* $\dim X = 0$*, then* $H^\bullet$ *is still supported in degree* 0*, and* $H^0(F)$ *is an étale algebra over* F*. Moreover,* $\int_X : H^0(X) \to F$ *is the "actual" trace.*

Definition 3.16 (Weil Cohomology Theory)**.** A *Weil Cohomology theory* is a pre-Weil cohomology theory such that the natural map

$$H^0(\mathrm{Spec}\ \Gamma(X, \mathcal{O}_X)) \to H^0(X)$$

is an isomorphism for each $X \in P(k)$.

Theorem 3.17. *If* $H^\bullet$ *is a Weil cohomology theory, and* A *is an abelian variety, then*

$$H^\bullet(A) \simeq \bigwedge^\bullet H^1(A)$$

4 Lefschetz Trace Formula and Applications

Let X be a smooth (geometrically irreducible) projective variety over $\mathbb{F}_q$. The Weil conjecture states some properties of

$$Z(X,t) = \exp(\sum_{n=1}^{\infty} \frac{t^n}{n} |X(\mathbb{F}_{q^n})|)$$

Today we will prove the rationality and functional equation of this function.

Rationality: we can write

$$Z(X,t) = \frac{P_1(t)P_3(t)\cdots P_{2d-1}(t)}{P_0(t)P_2(t)\cdots P_{2d}(t)}$$

for each $P_i \in 1 + t\mathbb{Z}[t]$.

Functional Equation:

$$Z(X, q^{-d}t^{-1}) = \pm q^{d\chi/2} t^{\chi} Z(X,t)$$

4.1 Recap of étale cohomology

Assume X/k for k algebraically closed with dimension d. We have two algebraic objects associated to X:

1. $\mathrm{CH}^*(X) = \bigoplus_{i=0}^{d} \mathrm{CH}^i(X)$ the Chow ring. It has the intersection product:

   $$\mathrm{CH}^i(X) \times \mathrm{CH}^j(X) \to \mathrm{CH}^{i+j}(X), \quad [A][B] = [A \cap B]$$

   for generically transverse intersection. We can extends uniquely to all of $\mathrm{CH}^*(X)$. We call A and B are generically transverse if for all $C \subset A \cap B$ irreducible component, a general point at $x \in C$ satisfies:

   $$T_x A + T_x B = T_x X$$

2. ℓ-adic cohomology ring

   $$H^*_\ell(X) = \bigoplus_{i=0}^{2d} H^i_\ell(X_{\text{ét}}, \mathbb{Q}_\ell)$$

   we have a cup product on this. This is a graded commutative ring.

   If $\mathrm{char}(k) \neq l$, then we have these properties as a Weil cohomology theory:

   (a) Künneth: $H^*(X \times Y) \simeq H^*(X) \times H^*(Y)$

   (b) Poincaré duality:

   $$H^i(X_{\text{ét}}, \mathbb{Q}_\ell) \times H^{2d-i}(X_{\text{ét}}, \mathbb{Q}_\ell(d)) \to H^{2d}(X_{\text{ét}}, \mathbb{Q}_\ell(d)) \xrightarrow{\mathrm{tr}} \mathbb{Q}_\ell$$

   (c) Cohomological dimension: for $i > 2d$

   $$H^i(X_{\text{ét}}, \mathbb{Q}_\ell) = 0$$

   (d) Cycle class map:

   $$\mathrm{cl}_X : \mathrm{CH}^* X \to \bigoplus H^i(X_{\text{ét}}, \mathbb{Q}_\ell(i))$$

   sending Z to $i_* 1_Z$. (We define $\mathrm{cl}_X X = 1_X \in H^0$).

4.2 Lefschetz Trace Formula

This is also called the fxied point theorem. Let $\varphi : X \to X$ be regular morphism. We wish to number of fixed points of φ. This should defined to be $\Gamma_\varphi \cdot \Delta$, where Γ_φ is the graph of φ and Δ is the diagonal. The Lefschetz Trace formula states:

$$\Gamma_\varphi \cdot \Delta = \sum_{r=0}^{2d} (-1)^r \mathrm{Tr}_{H^r}(\varphi^*)$$

Proof. Note

$$\mathrm{cl}_{X\times X}(\Gamma_\varphi \cdot \Delta) = \mathrm{cl}_{X\times X}(\Gamma_\varphi) \cdot \mathrm{cl}_{X\times X}(\Delta)$$

We take a basis $e_{i,r}$ for $H^*(X)$. Poincare duality will give me a dual basis $f_{i,2d-r}$ for $H^*(X)$, in the sense

$$e_{i,r} \cup f_{j,2d-r} = \delta_{ij} \cdot e_{2d}$$

By Künneth, we can write

$$\mathrm{cl}_{X\times X}(\Gamma_\varphi) = \sum_{i,r} a_{i,r} \otimes f_{i,2d-r}$$

where $a_{i,r} \in H^r$. So

$$\mathrm{cl}_{X\times X}(\Gamma_\varphi) \cup (1 \otimes e_{j,r}) = a_{j,r} \otimes e_{2d}$$

Let $p, q : X \times X \to X$ be the projections to the first and second factor. We pushforward the above expression via p:

$$a_{jr} = p_*(\mathrm{cl}_{X\times X}(\Gamma_\varphi) \cup q^* e_{j,r}) = p_*((1,\varphi)_* 1_X \cup q^* e_{j,r}) = \varphi^* e_{j,r}$$

Then:

$$\mathrm{cl}_{X\times X}(\Gamma_\varphi) = \sum_{j,r} \varphi^* e_{j,r} \otimes f_{j,2d-r}$$

and

$$\mathrm{cl}_{X\times X}(\Delta) = \sum_{j,r} e_{j,r} \otimes f_{j,2d-r} = \sum_{j,r} (-1)^r f_{j,2d-r} \otimes e_{j,r}$$

So we have:

$$\begin{aligned} \mathrm{cl}_{X\times X}(\Gamma_\varphi) \cdot \mathrm{cl}_{X\times X}(\Delta) &= \sum_{j,r} (-1)^r [(\varphi^* e_{j,r}) \cdot f_{j,2d-r}] \otimes e_{2d} \\ &= \sum_{r} (-1)^r \mathrm{Tr}_{H^r}(\varphi^*) e_{2d} \otimes e_{2d} \end{aligned}$$

The result follows. □

4.3 Proof of Rationality and Functional Equation

Note $X(\mathbb{F}_{q^n})$ are precisely the fixed points of Frob^n acting on $X(\overline{\mathbb{F}_q})$. We apply trace formula for $\varphi = \mathrm{Frob}^n$, we see

$$|X(\mathbb{F}_{q^n})| = \sum_{r=0}^{2d} (-1)^r \mathrm{Tr}_{H^r}(\mathrm{Frob}^n)^*$$

There is a potential gap: we need to check the graph of Frobenius and the diagonal intersects transversely. The idea is that the tangent spaces has trivial intersection and hence they span the whole tangent space $T_{(x,x)}(X \times X)$ (since the dimension is correct).

Suppose $\mathrm{Frob}^*_{H^r}$ has eigenvalues $\lambda_1, \cdots, \lambda_{h_r}$. So $\mathrm{Frob}^{*n}_{H^r}$ has eigenvalues $\lambda_1^n, \cdots, \lambda_{h_r}^n$, which implies $\mathrm{Tr}(\mathrm{Frob}^{*n}_{H^r}) = \lambda_1^n + \cdots \lambda_{h_r}^n$. Hence:

$$\begin{aligned}
Z(X,t) &= \exp\Big(\sum_{n=1}^{\infty} \frac{t^n}{n} \sum_{r=0}^{2d} (-1)^r (\lambda_1^n + \cdots \lambda_{h_r}^n)\Big) \\
&= \prod_{r=0}^{2d} \prod_{i=1}^{h_r} \exp\Big(\sum_{n=1}^{\infty} (-1)^r \frac{(t\lambda_i)^n}{n}\Big) \\
&= \prod_{r=0}^{2d} \prod_{i=1}^{h_r} (1 - t\lambda_i)^{(-1)^{r+1}} \\
&= \prod_{r=0}^{2d} \det(1 - t\mathrm{Frob}^*_{H_r})^{(-1)^{r+1}}
\end{aligned}$$

By renaming these terms, we get get the rationality.

We can also write

$$Z(X,t) = \prod_{x \in X \text{ closed points}} (1 - t^{\deg x})^{-1} \in 1 + t\mathbb{Z}[[t]]$$

This will imply $P_i(t) \in 1 + t\mathbb{Z}[t]$

Proof. (of functional equation). We know the degree of the Frobenius is q^d: $\mathrm{Frob}_* \mathrm{Frob}^* = q^d$. We need to compute:

$$\begin{aligned}
Z(X, q^{-d}t^{-1}) &= \prod_{r=0}^{2d} \det(1 - q^{-d}t^{-1}\mathrm{Frob}^*_{H_r})^{(-1)^{r+1}} \\
&= \prod_{r=0}^{2d} \det(1 - q^{-d}t^{-1}\mathrm{Frob}_{*,H_r})^{(-1)^{r+1}} \\
&= \prod_{r=0}^{2d} \det(1 - t^{-1}(\mathrm{Frob}^*_{H_r})^{-1})^{(-1)^{r+1}} \\
&= (-t)^{\chi} \prod_{r=0}^{2d} \det(1 - t\mathrm{Frob}^*_{H_r})^{(-1)^{r+1}} \det((\mathrm{Frob}^*_{H_r})^{-1})^{(-1)^{r+1}} \\
&= \pm t^{\chi} q^{d\chi/2} Z(X,t)
\end{aligned}$$

where $\chi = \sum(-1)^r \dim H^r$ is the Euler characteristic. $\square$

5 Constructible Sheaves, Base Change Theorem, L-functions

Here, we assume:

1. all schemes are finite type and separated over a field.
2. $(X_0, \mathcal{F}_0)$ are over finite field $\mathbb{F}_q$; $(X, \mathcal{F}) = (X_0 \otimes_{\mathbb{F}_q} \overline{\mathbb{F}_q}, \mathcal{F}_0 \otimes_{\mathbb{F}_q} \overline{\mathbb{F}_q})$
3. Fix an invertible prime ℓ.

5.1 Constructible Sheaf

Theorem 5.1. *If $f : X \to Y$ smooth proper, $\mathcal{L}$ local system on X. Then for all $i \geq 0$, we have:*

$$R^i f_* \mathcal{L}$$

is a local system on Y.

Remark. *If X, Y are real manifolds, then Ehresmann's lemma states that f is a locally trivial fibration.*

Question: What if f is not proper or not smooth?

Definition 5.2. Let $\mathcal{F}$ be a sheaf on X with finite stalks (coprime to p). We call $\mathcal{F}$ constructible if there is a locally closed decomposition $X = \bigcup X_i$ such that $\mathcal{F}|_{X_i}$ is a local system.

Example. *Let $i : Z \to X$ be closed, then $i_* \underline{\mathbb{Z}/\ell^n}$ is constructible.*

Theorem 5.3. *Let $f : X \to Y$ be any morphism, $\mathcal{F}$ constructible on X. Then for all $i \geq 0$, $R^i f_* \mathcal{F}$ are constructible.*

Theorem 5.4. *Consider $U \overset{j}{\hookrightarrow} X \overset{i}{\hookleftarrow} Z$, for j open embedding and $Z = X \backslash U$. Then the data of constructible sheavf on X is the same as the data of a triple $(\mathcal{F}_1, \mathcal{F}_2, f)$ where:*

- *$\mathcal{F}_1$ constant sheaf on Z*
- *$\mathcal{F}_2$ constructible sheaf on U*
- *$f : \mathcal{F}_1 \to i^* j_* \mathcal{F}_2$*

Example. *Let X be a smooth curve over $\mathbb{C}$, use real topology. Let's give a constructible sheaf on X. We set $U = X \backslash Z$ for $Z \subset X$ finite, and $\mathcal{L}$ a local system on U: $\pi_1(U)$-module L. For any $z \in Z$, we need to give a finite $\mathbb{Z}/\ell^n$-module V_z. Finally, for any $z \in Z$, we need a morphism $V_z \to L^{I_z}$, where $I_z \simeq \mathbb{Z}$ is a loop generator of monodromy near z.*

Example. *Let $X_0/\mathbb{F}_q$. Then the data of constructible sheaf on X_0 i the same as:*

- *$U_0 \subset X_0$ open, $Z_0 = X_0 \backslash U_0$*
- *$\mathcal{L}$ local system on U_0*
- *For any $z \in Z_0$, need a $\mathrm{Gal}(\overline{k(z)}/k(z))$-module V_z.*
- *$V_z \to L^{I_z}$ of Galois modules.*

where I_z is the inertia group.

Definition 5.5 (ℓ-adic Sheaf)**.** An ℓ-adic sheaf is a compatible system of $\mathbb{Z}/\ell^n$-construtible sheaves. In particular, all stalks are finite dimensional finite timensional $\mathbb{Q}_\ell$-vector space.

5.2 Proper Base Change

$$\begin{array}{ccc} \widetilde{X} & \xrightarrow{\widetilde{f}} & X \\ \downarrow{\scriptstyle \widetilde{f}} & & \downarrow{\scriptstyle f} \\ \widetilde{Y} & \xrightarrow[g]{} & Y \end{array}$$

Let $\mathcal{F}$ be an ℓ-adic sheaf on X. There is a "base change" morphism:

$$g^* f_* \mathcal{F} \to \widetilde{f}_* \widetilde{g}^* \mathcal{F}$$

and more generally

$$g^* R^i f_* \mathcal{F} \to R^i \widetilde{f}_* \widetilde{g}^* \mathcal{F}$$

These are not isomorphisms in general, even if everything is Cartesian.

Example.

$$\begin{array}{ccc} \emptyset & \longrightarrow & \mathbb{G}_m \\ \downarrow & & \downarrow{\scriptstyle f} \\ 0 & \xrightarrow[g]{} & \mathbb{A}^n \end{array}$$

Let $\mathcal{F} = \underline{\mathbb{Q}}_\ell$, but then $g^* f_* \underline{\mathbb{Q}}_\ell \simeq \mathbb{Q}_\ell$.

Theorem 5.6 (proper base change)**.** *If the above diagram is Cartesian, and f is proper, then*

$$g^* R^i f_* \mathcal{F} \to R^i \widetilde{f}_* \widetilde{g}^* \mathcal{F}$$

is an isomorphism.

5.3 Cohomology with Compact Support

Let $j : U \to X$ be open embedding, $\mathcal{F}$ a sheaf on U. Let $i : Z = X \backslash U \to X$ be closed embedding.

Theorem 5.7. *There exists a functor $j_! : \mathrm{Shv}(U) \to \mathrm{Shv}(X)$ is left adjoint to j^**

Remark. *1. $j_!$ is exact*

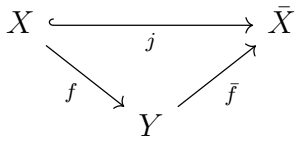
2. $(j_!\mathcal{F})_x = \begin{cases} \mathcal{F}_x, & \text{if } x \in U; \\ 0, & \text{otherwise.} \end{cases}$

3. For any sheaf $\mathcal{F}$ oon X, we have an exact sequence:

$$0 \to j_! j^* \mathcal{F} \to \mathcal{F} \to i_* i^* \mathcal{F} \to 0$$

Let $f : X \to Y$ any morphism.

Theorem 5.8 (Nagata)**.** *There exists a compactification*



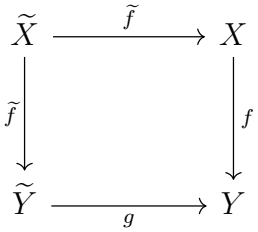
such that j open embedding, and $\overline{f}$ is proper.

Definition 5.9. Let $f \in \mathrm{Shv}(X)$. We define:

$$R^i f_! \mathcal{F} \coloneqq R^i \overline{f}_* (j_! \mathcal{F})$$

Remark. *1. $R^i f_!$ preserves constructibility.*

2. If



is Cartesian. $g^* R^i f_!(\mathcal{F}) = R^i \widetilde{f}_!(\widetilde{g}^* \mathcal{F})$

3. We can define $H_c^i(X, \mathcal{F}) \coloneqq R^i p_! \mathcal{F}$ for $p : X \to \mathrm{Spec}\ \overline{k}$.

4. $R^i f_!$ are **not** *derived functors of $f_!$.*

5.4 Frobenius

Let X_0/k for k finite field, $\mathcal{F}_0 \in \mathrm{Shv}(X_0)$. There is a $\mathrm{Gal}(\overline{k}/k)$-action on $H^i(X, \mathcal{F})$, and on $H^i_c(X, \mathcal{F})$ by "transport of structure".

Given $\sigma : \mathrm{Gal}(\overline{k}/k)$, we get:

$$H^i(X, \mathcal{F}) \xrightarrow{(\sigma \times 1_{X_0})^*} H^i(X, (\sigma \times 1_{X_0})^* \mathcal{F}) \simeq H^i(X, \mathcal{F})$$

We have the following convention: we write $\mathrm{Fr}^{\mathrm{arith}} \in \mathrm{Gal}(\overline{k}/k)$ for $x \mapsto x^{|k|}$, and $\mathrm{Fr} \coloneqq (\mathrm{Fr}^{\mathrm{arith}})^{-1}$.

5.5 L-functions of ℓ-adic Sheaves

Definition 5.10. Given $(X_0 \to \mathrm{Spec}\ k, \mathcal{F}_0)$, we define:

$$Z(X_0/\mathrm{Spec}\ k, \mathcal{F}_0, T) = \prod_{x \in |X_0|} \frac{1}{\det(1 - \mathrm{Fr}_x \cdot T^{\deg x}, \mathcal{F}_x)}$$

Proposition 5.11. *We have*

$$Z(X_0/\mathrm{Spec}\ k, \mathcal{F}_0, T) = \exp(\sum_{m=1}^{\infty} N_m \frac{T^m}{m})$$

where $N_m = \sum_{x \in X_0(k_m)} \mathrm{Tr}(\mathrm{Fr}_x, \mathcal{F}_x)$

Theorem 5.12 (Grothendieck-Lefschetz Trace Formula)**.** *Given $(X_0, \mathcal{F}_0)$, the following equality holds:*

$$Z(X_0/k, \mathcal{F}_0, T) = \prod_{i=0}^{2 \dim X} \det(1 - \mathrm{Fr}_k \cdot T, H^i_c(X, \mathcal{F}))^{(-1)^{i+1}}$$

Definition 5.13. Given $(X_0, \mathcal{F}_0)$, let $G_{\mathcal{F}_0} : X_0(k_m) \to \overline{\mathbb{Q}}_l$ sending $x \mapsto \mathrm{Tr}(\mathrm{Fr}_x, \mathcal{F}_x)$

Theorem 5.14 (Grothendieck-Lefschetz Trace Formula)**.** *For any $f_0 : X_0 \to Y_0$, $\mathcal{F}_0 \in \mathrm{Shv}(X_0)$, $m \geq 1$. We have*

$$K_0(\mathrm{Shv}(X_0)) \xrightarrow{G} \mathrm{Func}(X(k_m), \mathbb{Q}_\ell)$$

for K_0 the Grothendieck group of sheaves. We also have:

$$K_0(\mathrm{Shv}(X_0)) \to K_0(\mathrm{Shv}(Y_0))$$

sending $\mathcal{F}$ to $\sum_i (-1)^i (R^i f_! \mathcal{F})$. Then if we identify both sides using G, the map is simply integrating along fibres.

6 Katz's proof of the Riemann hypothesis for curves

Fix genus g and prime q. Suppose we know Riemann hypothesis holds for one curve of genus g: $C_0/\mathbb{F}_q$. We wish to extend this knowledge to other curves via the moduli space of genus g curves.

6.1 Review

Let X be a scheme of finite type over $\mathbb{F}_q$. Recall the zeta function:

$$Z(X,T) = \exp(\sum_{n\geq 1} \#X(\mathbb{F}_{q^n})T^n/n) \in \mathbb{Q}[[T]]$$

Now if C is a smooth proper curve of genus g, we have:

$$Z(C,T) = \frac{P(C,T)}{(1-T)(1-qT)}$$

for $P(C,T) = 1 + C_1T + C_2T^2 + \cdots C_{2g}T^{2g} = \prod_i(1-\alpha_iT)$. The Riemann Hypothesis states $|\alpha_i| = q^{\frac{1}{2}}$.

We denote $F_q \in \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ to be the geometric Frobenius. Let

$$P_i(X,T) = \det(1 - TF_q|_{H^i(\overline{X},\overline{\mathbb{Q}}_\ell)})$$

Let $\mathcal{F}$ be ℓ-adic local system on U_0. This is a finite-dimensional $\pi_1^{\text{ét}}(U_0)$ representation. So given a closed point $\mathrm{Spec}\ k \to U_0$, one can define a Frobenius $\mathrm{Frob}_\mathfrak{p}|_\mathcal{F}$ up to conjugacy. So $\det(1 - T\mathrm{Frob}_\mathfrak{p}|_\mathcal{F})$ is well-defined.

If we set $\mathcal{F} = R^if_*\mathbb{Q}_\ell$. Proper base change tells as that at closed point $\mathfrak{p}$:

$$\det(1 - T^{\deg\mathfrak{p}}\mathrm{Frob}_\mathfrak{p}|R^if_*\mathbb{Q}_\ell) = \det(1 - (TF_q)^{\deg\mathfrak{p}}|H^i_{\text{et}}(\overline{X}_\mathfrak{p}, \overline{\mathbb{Q}_\ell}))$$

Use this we can manipulate the L-function to have form:

$$Z(U_0,\mathcal{F},T) = \frac{\det(1 - TF_q|H^1_c(U,\mathcal{F}))}{\det(1 - qTF_q|(\mathcal{F})_{\pi_1^{\text{geom}}})}$$

6.2 Reduction to Single Curve

Definition 6.1. Let $\iota : \overline{\mathbb{Q}}_\ell \to \mathbb{C}$. We fix ι and write $|\beta| := |\iota(\beta)|$. Let $\mathcal{F}$ be an ℓ-adic local system on U_0, where U_0 is affine smooth geometrically connected curve. We say $\mathcal{F}$ is **ι-pure of weight** $w \in Z$ if:

$$|\alpha_i| = N(\mathfrak{p})^{\frac{w}{2}}$$

for all closed points $\mathfrak{p} \in U_0$ and all eigenvalues α of $\mathrm{Frob}_\mathfrak{p}|_\mathcal{F}$

We say $\mathcal{F}$ is **ι-real** if

$$\det(1 - T\mathrm{Frob}_\mathfrak{p}|\mathcal{F}) \in \mathbb{R}[x]$$

for all closed $\mathfrak{p} \in U_0$

Theorem 6.2. *Let $\mathcal{F}$ be an ι-real ℓ-adic local system on U_0. Suppose for every $\mathcal{F}^{\otimes 2k}$, we have every eigenvalue β_{2k} of $F_q|_{(\mathcal{F}^{\otimes 2k})_{\pi_1^{\mathrm{geom}}}}$ has $|\beta_{2k}| \leq 1$. Then for every closed point $\mathfrak{p}$, we have every eigenvalue $\alpha_{i,\mathfrak{p}}$ of $\mathrm{Frob}_{\mathfrak{p}}|_{\mathcal{F}}$ has $|\alpha_{i,\mathfrak{p}}| \leq 1$.*

Corollary 6.3. *If there exists a closed point $\mathfrak{p}_0$ such that all eigenvalues of α of $\mathrm{Frob}_{\mathfrak{p}_0}|_{\mathcal{F}}$ has $|\alpha| \leq 1$, then all closed points have this property.*

Proof. (Sketched) Let $d = \deg \mathfrak{p}_0$. Then on $(\mathcal{F}^{\otimes 2k})_{\pi_1^{\mathrm{geom}}}$, F_q^d acts as $\mathrm{Frob}_{\mathfrak{p}_0}$. □

We say "nice" for smooth projective, geometrically connected; "nice affine" for smooth affine geometrically connected.

Lemma 6.4. *Let $g \geq 1$, and C_0, C_1 are two nice curves over $\mathbb{F}_q$ of genus g. After base change to $E/\mathbb{F}_q$, there exists U_0 "nice affine" curve and proper smooth morphism $f : C \to U_0$ such that C_1, C_0 are its fibres.*

Proof. If $g = 1$, pick $N \geq 4$. After base change, we may assume C_0 and C_1 has point of order N. □

Theorem 6.5. *Let C_0, C_1 be nice curves over $\mathbb{F}_q$ with genus $g \geq 1$. Then the RH for C_1 implies RH for C_0.*

6.3 RH Holds for One Curve

Definition 6.6. Let $d \geq 2$, d coprime to q. Let F_d to be the Fermat curve:

$$x^d + y^d = z^d$$

over $\mathbb{P}^2_{\mathbb{F}_q}$

Theorem 6.7 (Weil)**.** *RH holds for all Fermat curves.*

The idea is the proof is to using the Jacobi sum. Given $x_1, x_2 : \mathbb{F}_q^\times \to \mathbb{C}^\times$, we define:

$$J(x_1, x_2) = \sum_{\alpha \in \mathbb{F}_q} x_1(\alpha) x_2(1 - \alpha)$$

with $x(0) = 0$. It has the property that if $x_1, x_2, x_1 \cdot x_2$ are nontrivial, then $|J(x_1, x_2)| = q^{\frac{1}{2}}$.

Note F_d has genus $\binom{d-1}{2}$. We want to get all genus.

Lemma 6.8. *Let $g \geq 1$, then there exists $d \geq 2$ coprime to q such that $F_d/\mathbb{F}_q$ has "nice" quotient C of genus g.*

Proof. We can construct curves of form $x^d + y^2 = 1$ for $p \neq 2, d = 2g + 1$ or $2g + 2$; and curve $y^2 - y + x^{2g+1} = 0$ for $p = 2$. □

6.4 Persistence of Purity

Let $\mathcal{F}$ be a ι-real, ℓ-adic local system on $U - 0$. If there exists a closed $\mathfrak{p}_0$ such that every eigenvalue $\alpha_{0,\iota}$ of $\mathrm{Frob}_{\mathfrak{p}_0}|_{\mathcal{F}}$ has $\iota(\alpha_{0,\iota}) = 1$ then the same condition holds for all closed point $\mathfrak{p}$. (This say $\mathcal{F}$ is ι-pure of weight 0).

7 Riemann Hypothesis for Hypersurfaces

Let $U_0/\mathbb{F}_q$, $\mathcal{F}$ an ℓ-adic local system on U_0. Let $u \in U_0$ a closed point, we can define $\mathrm{Frob}_u \in \pi_1(U_0)$ defined up to conjugacy. So $\det(1 - T\mathrm{Frob}_u|\mathcal{F})$ is well-defined.

Suppose I have $f : X \to U_0$ smooth proper family, $\mathcal{F} = R^i f_* \underline{\mathbb{Q}_\ell}$.

$$\det(1 - T\mathrm{Frob}_u|R^i f_*\underline{\mathbb{Q}_\ell}) = \det(1 - T\mathrm{Frob}_{q^{\deg(u)}} \mid H^i(X_u, \underline{\mathbb{Q}_\ell})) = P_i(X_u, T)$$

$$Z(X_u/\mathbb{F}_{q^{\deg u}}, T) = \prod_{i=0}^{2n} P_i(X_u, T)^{(-1)^{i+1}}$$

FIx $\iota : \overline{\mathbb{Q}_\ell} \to \mathbb{C}$.

Corollary 7.1. *1. $P_i(X_u, T) \in 1 + T\mathbb{Z}[T] \implies R^i f_*\underline{\mathbb{Q}_\ell}$ is real.*

2. RH holds for $X_u \iff H^i(X_u, \mathbb{Q}_\ell)$ are pure of weight $i \iff R^i f_*\underline{\mathbb{Q}_\ell}$ is pure of weight i at u.

Theorem 7.2 (Persistence of Purity)**.** *Let U_0 be affine smooth geometrically connected curve over $\mathbb{F}_q$. Let $\mathcal{F}$ be an ℓ-adic local system on U_0 which is real. Then $\mathcal{F}$ is pure of weight 0 at some $u \in U_0$, then $\mathcal{F}$ is pure of weight zero on U_0.*

Definition 7.3 (Tate twists)**.** 1. $\mathbb{Q}_\ell(i)$ is an one-dimensional $\mathbb{Q}_\ell$ vector space with an action of Frob_q as q^{-i}.

2. If V is equipped with a Frobenius action, then we define $V(i) = V \otimes \mathbb{Q}_\ell(i)$

3. We can twist any local system.

Clearly, if V is pure of weight w (namely $|\alpha| = q^{\frac{w}{2}}$), then $V(i)$ is pure of weight $w - 2i$.

Let's discuss cohomology of cohomology of $\mathbb{P}^n$.

$$H^{2i+1}(\mathbb{P}^n, \mathbb{Q}_\ell) = 0; \quad H^{2i}(\mathbb{P}^n, \mathbb{Q}_\ell) = \mathbb{Q}_\ell(-i)$$

Here is a sanity check:

$$Z(\mathbb{P}^n, T) = \prod_{i=0}^{n} \frac{1}{\det(1 - T\mathrm{Frob}_q|\mathbb{Q}_\ell(-i))} = \prod_{i=0}^{n} \frac{1}{\det(1 - Tq^i|\mathbb{Q}_\ell)} = \prod_{i=0}^{n} \frac{1}{1 - q^i T}$$

this agrees with the point counts.

Let $X \subset \mathbb{P}^{n+1}$ smooth of degree d defined over $\mathbb{F}_q$. Weak Lefschetz theorem and Poincaré duality implies

$$H^i(X, \mathbb{Q}_\ell) = H^i(\mathbb{P}^{n+1}, \mathbb{Q}_\ell), i \neq n$$

and

$$H^n(X, \mathbb{Q}_\ell) \supset H^n(\mathbb{P}^{n+1}, \mathbb{Q}_\ell)$$

Let's define

$$\mathrm{Prim}^n(X) = \begin{cases} H^n(X, \mathbb{Q}_\ell), & \text{if } n \text{ odd;} \\ H^n(X, \mathbb{Q}_\ell/\langle L^{\frac{n}{2}} \rangle), & \text{otherwise.} \end{cases}$$

and

$$P(T) = \det(1 - T\mathrm{Frob}_q | \mathrm{Prim}^n(X))$$

Hence:

$$Z(X,T) = \begin{cases} P(T)Z(\mathbb{P}^n, T), & \text{if } n \text{ odd;} \\ \dfrac{1}{P(T)} Z(\mathbb{P}^n, T), & \text{otherwise.} \end{cases}$$

Corollary 7.4. *RH holds for $X \iff \mathrm{Prim}^n(X)$ is pure of weight $n \iff H^n(X, \mathbb{Q}_\ell)$ is pure of weight n.*

Theorem 7.5. *Fix (n, d, p). If there exists a smooth hypersurface $X_0 \subset \mathbb{P}^{n+1}$ of degree d, defined over $\mathbb{F}_p$ such that RH holds for it. Then for any $\mathbb{F}_q/\mathbb{F}_p$ and any $X_1 \subset \mathbb{P}^{n+1}$ of degree d, defined over $\mathbb{F}_q$, Riemann hypothesis holds for X.*

Proof. Consider $X_0 \otimes \mathbb{F}_q$, RH holds for it. So we can assume $X_0/\mathbb{F}_q$. Now $\mathbb{P}^{n+1} \subset X_0, X_1$ defined by F_0 and F_1. Consider $tF_1(x)+(1-t)F_0(x)$, this is a family over $\mathbb{A}^1$ {finite set} $= U_0$. Then $R^n f_* \underline{\mathbb{Q}_\ell}$ is pure of weight n at 0. By extending to $\overline{\mathbb{Q}_\ell}$, we can make a twist and make the local system be pure of weight 0at 0. By persistence of purity, the local system is pure of weight 0 on U_0, hence RH holds for X_1. □

Remark. *1. By Poincaré duality, we need to prove that $|\alpha| \leq q^{\frac{n}{2}}$ for α eigenvalue of Frob_q-action on $H^n(X, \mathbb{Q}_\ell)$*

2. If X is defined over $\mathbb{F}_q$, $q = p^r$, then

$$\#X(\mathbb{F}_q) = \#\mathbb{P}^n(\mathbb{F}_q) + (-1)^n \mathrm{Tr}(\mathrm{Frob}_q \mid \mathrm{Prim}^n(X, \mathbb{Q}_\ell))$$

So it suffices to show $\#X(\mathbb{F}_q) = \#\mathbb{P}^n(\mathbb{F}_q) + O(q^{\frac{n}{2}})$.

Here is the the list of the testing hypersurfaces X:

1. If $p \nmid d$, consider $\sum_{i=0}^{n+1} x_i^d = 0$ the Fermat's hypersurface.
2. If $p \nmid d, d \geq 3$, consider $x_1^d + \sum_{i=1}^{n+1} x_i x_{i+1}^{d-1} = 0$ the Gabber's hypersurface
3. If $p = d = 2$, n odd, then $\mathrm{Prim}^n = 0$, we're done.
4. If $p = d = 2$, $n = 2m$, consider $\sum_{i=1}^{m+1} x_i x_{m+i+1} = 0$

Reduce to the affine case, set $V = X^{\mathrm{aff}}$. We also rename $N = n + 2$. Need to show:

$$\#V(\mathbb{F}_q) = q^{n+1} + O(q^{\frac{n+2}{2}}) = q^{N-1} + O(q^{\frac{N}{2}})$$

Let's define $V^*(\mathbb{F}_q) = \{x \in \mathbb{G}_m^N(\mathbb{F}_q) \mid \sum x_i^d = 0\}$.

Theorem 7.6. $\#V^*(\mathbb{F}_q) = \frac{(q-1)^N}{q} + O(q^{\frac{N}{2}})$

By a simple calculation we see this implies our desired estimate.

Multiplicative characters: $\mathbb{F}_q^\times \to \mathbb{C}^\times$. $(\mathbb{F}_q^\times)^\vee \simeq \mu_{q-1}$. For example, the Legendre symbol is a multiplicative character.

Additive characters: $\mathbb{F}_q \to \mathbb{C}^\times$. We see $\psi(t) = \zeta_p^{\mathrm{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(t)}$. In fact, all characters has form $\psi_a(t) = \psi(at)$

Definition 7.7 (Gauss sums)**.** Let ψ_a be an additive character, $a \neq 0$. Let χ be a multiplicative character.

$$g\psi_a(\chi) = \sum_{t \in \mathbb{F}_q^\times} \chi(t)\psi_a(t)$$

We know

$$|g\psi_a(\chi)| = \begin{cases} \sqrt{q}, & \text{if } \chi \neq 1; \\ 1, & \text{if } \chi = 1. \end{cases}$$

Now consider $\varphi : \mathbb{G}_m^N(\mathbb{F}_q) \to \mathbb{G}_m^N(\mathbb{F}_q), (x_i) \mapsto (x_i^d)$, and $\sigma : \mathbb{G}_m^N(\mathbb{F}_q) \to \mathbb{A}^1, (t_i) \mapsto \sum t_i$. Then $V^*(\mathbb{F}_q) = \{x \in \mathbb{G}_m^N(\mathbb{F}_q) | \sigma(\varphi(x)) = 0\}$.

We see $\#V^*(\mathbb{F}_q) = \#\ker(\varphi) \cdot \#\{t \in \mathbb{G}_m^N(\mathbb{F}_q) \mid t \in \text{im }(\varphi), \sum t_i = 0\}$.

We see

$$\mathbf{1}_{\sum t_i = 0} = \frac{1}{q} \sum_{a \in \mathbb{F}_q} \psi_a(\sum t_i)$$

$$\mathbf{1}_{t \in \text{im }(\varphi)} = \frac{1}{\#\text{coker }(\varphi)} \sum_{\chi \in \text{coker }(\varphi)^\vee} \chi(t)$$

hence:

$$\#V^*(\mathbb{F}_q) = \frac{1}{q} \sum_{a \in \mathbb{F}_q} \sum_{\chi \in \text{coker}^\vee} \sum_{t \in \mathbb{G}_m^N(\mathbb{F}_q)} \chi(t)\psi_a(\sum_{i=1}^N t_i)$$

The term for $a = 0$ gives the leading term $\frac{1}{q}(q-1)^N$. We only need to then bound the case for $a \neq 0$.

8 Deligne's Proof Part I

8.1 Étale Fundamental Group

Let X/k be connected, finite type. Fix $\overline{x} \in X(\overline{k})$.

Definition 8.1. The fibre functor is

$$\omega_{\overline{x}} : \mathsf{FEt}(X) \to \mathsf{Set}$$

sending $Y \mapsto Y_{\overline{x}}$. We also define $\pi_1(X, \overline{x}) = \text{Aut}(\omega_{\overline{x}})$. So $\omega_x : \mathsf{FEt}(X) \to \pi_1(X, \overline{x}) - \mathsf{Set}$

Theorem 8.2. *There is a cofiltered sysmtem $X_i \in \mathsf{FEt}(X)$ such that*

$$\omega_{\overline{x}}(Y) = \operatorname*{colim}_i \text{Hom}_X(X_i, Y)$$

Moreover, we can take X_i Galois over X. Hence $\pi_1(X, \overline{x}) \simeq \lim \text{Aut}_X(X_i)$

Example. *Let $X = \text{Spec } k$. Take $X_i = \text{Spec } L$. We see $\pi_1(X, \overline{x}) = \text{Gal}(\overline{k}/k)$.*

Example. *Consider* $X = \mathbb{G}_m$, *and* $\mathbb{G}_m \xrightarrow{z \mapsto z^n} \mathbb{G}_m$ *for* $\mathrm{char} k = 0$ *and* $k = \overline{k}$. *We get:*

$$\pi_1(X, \overline{x}) = \hat{\mathbb{Z}}$$

If fact for $X/\mathbb{C}$, *we have:*

$$\pi_1^{\text{ét}}(X, \overline{x}) \simeq \hat{\pi_1}^{\text{top}}(X(\mathbb{C}), \overline{x})$$

Let $\mathrm{Loc}(X, \mathsf{Set})$ be the locally constant finite set-valued sheaves on $X_{\text{ét}}$. We can extend $\omega_{\overline{x}}$ to

$$\mathrm{Loc}(X, \mathsf{Set}) \to \pi_1\text{-}\mathsf{Set}$$

Theorem 8.3. *1. This is an equivalence of categories.*

2. All such sheaves are representatble.

We can extend ω_X to

$$\mathrm{Loc}(X_{\text{ét}}, \mathbb{Q}_\ell) \to \mathrm{Rep}(\pi_1, \mathbb{Q}_\ell)$$

where the right hand side we mean continuous representations. This is again an equivalence of categories.

Lemma 8.4. *Let* $\mathcal{F} \in \mathrm{Loc}(X_{\text{ét}}, \mathbb{Q}_\ell)$, *then:*

$$H^0(X_{\text{ét}}, \mathcal{F}) \simeq (\mathcal{F}_{\overline{x}})^{\pi_1(X, \overline{x})}$$

Proof. The statement is clearly true for constant sheaf $\mathcal{F}$. Otherwise, Zariski locally over some cover $\widetilde{X}$ it is constant, and compute use Čech cohomology (all input for this step is from π_1). □

8.2 Main Lemma

Let $X_0/\mathbb{F}_q$. Fix $\mathcal{F}_0 \in \mathrm{Loc}(X_{\text{ét}}, \mathbb{Q}_\ell)$. Again as before we have base change $X, \mathcal{F}$ to $\overline{\mathbb{F}}_q$
Recall we showed

$$Z_{X_0, \mathcal{F}_0}(t) = \prod_i \det(1 - t\mathrm{Fr}, H_c^i(X_{\text{ét}}, \mathcal{F}))^{(-1)^{i+1}}$$

For an affine curve, the only contribution is from H_c^1 and H_c^2.

Fix an affine curve $U_0/\mathbb{F}_q = \mathbb{P}^1 \backslash S$ for S finite set of closed points. Fix $\mathcal{F}_0 \in \mathrm{Loc}(U_0, \mathbb{Q}_\ell)$. Fix $u \in |U|$ geometric point.

We say $\mathcal{F}$ is of weight β if for all $u_0 \in |U_0|$, the eigenvalue of Fr_{U_0} acting on $\mathcal{F}|_{u_0}$

- are algebraic numbers over $\mathbb{Q}$
- Complex conjugates have absolute values $q^{\deg u_0 \cdot \beta/2}$

Theorem 8.5 (Main Lemma)**.** *Assume*

1. we have a sympectic pairing

$$\psi : \mathcal{F}_0 \otimes \mathcal{F}_0 \to \mathbb{Q}_\ell(-\beta)$$

2. *Big monodromy. The image of* $\pi_1(U,u)$ *in* $\mathrm{GL}(\mathcal{F}_u)$ *lies in* $\mathrm{Sp}(\mathcal{F}_u,\psi_u)$ *and is open (in* ℓ*-adic topology).*

3. *(Rationality) characteristic polynomial of* Fr_{u_0} *acting on* $\mathcal{F}_{u_0}$ *is in* $\mathbb{Q}[t]$ *for a'l* $u_0 \in |U_0|$

Then $\mathcal{F}$ *has weight* β*.*

Example. *Suppose* $\pi : Y_0 \to U_0$ *family of smooth projective hypersurfaces in* $\mathbb{P}^{d+1}$ *for* d *odd. Take* $\mathcal{F}_0 = R^d\pi_*\mathbb{Q}_\ell$*. Poincare duality gives me*

$$R^d\pi_*\mathbb{Q}_\ell \times R^d\pi_*\mathbb{Q}_\ell \to R^{2d}\pi_*\mathbb{Q}_\ell = \mathbb{Q}_\ell(-d)$$

Rationality follows from Weil conjecture for hypersurfaces. If big monodromy, then Riemann Hypothesis follows.

Lemma 8.6. *If I have a symplectivity and big monodromy, then* $H^2_c(U,\mathcal{F}^{\otimes 2k}) \simeq \mathbb{Q}_\ell(-k\beta-1)^{\oplus N}$

Proof.

$$H^2_c(U,\mathcal{F}^{\otimes 2k}) \simeq H^0(U,\mathcal{F}^{\vee\otimes 2k})^\vee(-1) \simeq ((\mathcal{F}_u^{\vee\otimes 2k})^{\pi_1})^\vee(-1)$$

Let $G = \mathrm{Sp}(\mathcal{F}_u,\psi_u)$. It's a general fact that

$$\mathrm{Hom}(\mathcal{F}_u^{\otimes 2k},\mathbb{Q}_\ell) = \mathbb{Q}_\ell(k\beta)$$

The question is whether π_1-invariant is the same as G-invariant.

Let H be the largest subgroup of G sucht that fiexs $\mathrm{Hom}(\mathcal{F}_u^{\otimes 2k},\mathbb{Q}_\ell)^{\pi_1}$. We know H is Zariski-closed in G. But $\dim H = \dim G$, hence $H = G$ since G is connected. □

Here comes the Rankin's trick. Examine the 2 form of the zeta functions, we see the radius of convergence of $Z_{X_0,\mathcal{F}_0}$ is smaller than the termwise radius convergence of each term. This gives me:

$$\frac{1}{q^{k\beta+1}} \le \frac{1}{|\alpha|^{\frac{2k}{\deg u_0}}}$$

Hence $|\alpha| \le q_{u_0}^{\beta/2+\frac{1}{2k}}$. Take $k \to \infty$, we see $\alpha \le q_{u_0}^{\beta/2}$. But $\mathcal{F}_0$ is symplectic, so $q_{u_0}^\beta/\alpha$ is an eigenvalue. From this we deduce the main lemma.

8.3 Consequences

Let $j : U_0 \to \mathbb{P}^1/\mathbb{F}_q$. We want to study $H^1(\mathbb{P}^1, j_*\mathcal{F})$. Poincare duality induces perfect pairing:

$$H^1(\mathbb{P}^1,j_*\mathcal{F}) \otimes H^1(\mathbb{P}^1,j_*\mathcal{F}) \to \mathbb{Q}_\ell(-1)$$

Corollary 8.7. *Let* α *be an eigenvalue of* $H^1(\mathbb{P}^1,j_!\mathcal{F}) = H^1_c(U,\mathcal{F})$*, then:*

$$|\alpha| \le q^{\frac{\beta+1}{2}+\frac{1}{2}}$$

If I examine $H^1(\mathbb{P}^1,j_*\mathcal{F})$*, then*

$$q^{\frac{\beta+1}{2}-\frac{1}{2}} \le |\alpha| \le q^{\frac{\beta+1}{2}+\frac{1}{2}}$$

9 Deligne's Proof Part II

Assume $n = 2m + 1$ be odd.

9.1 Motivation

Theorem 9.1 (Ehresmann's Theorem)**.** *Let $f : Y \to S$ be smooth and proper, and $\mathcal{F}$ is a locally constant constructible sheaf on $Y_{\text{ét}}$, and $\mathcal{F}$ is invertible on S, then $R^p f_* \mathcal{F}$ remains to be locally constant constructible sheaf, and commutes with base change.*

Theorem 9.2 (Morse Lemma)**.** *Let M be* $\dim = d$ *closed manifold. There exists $U \subset \mathrm{Fun}(M, \mathbb{R})$ in C^2-topology such that:*

- *There exists finitely many critical points, at most one in each fibre.*
- *Cirtical points are nondegenerate. Moreover, p critical, $f^{-1}(p)$ is described as $0 = x_1^2 + \cdots x_i^2 - x_{i+1}^2 - \cdots - x_d^2$.*

9.2 Lefschetz Pencil

Definition 9.3. Let $|\mathcal{L}|$: $X \to \mathbb{P}^N$ be very ample line bundle. A **Lefschetz pencil** D for $(X, \mathcal{L})$ is a one-parameter line of hyperplanes $D \subset \mathbb{P}^{\vee N}$. We can write:

$$D = \{H_t \subset \mathbb{P}^N\}_{t \in \mathbb{P}^1}$$

The axis of a pencil is

$$A \coloneqq \bigcap_t H_t = H_0 \cap H_\infty$$

We require:

1. A intersects X transversely
2. The intersections $X_t = X \cap H_t$ are nonsingular for all but finitely many t
3. The bad set of time $S \subset \mathbb{P}^1$, $t \in S$, the singularity is a single ordinary double point.

Remark. *An ordinary double point $p \in X/k$ is a geometric point such that*

$$\hat{\mathcal{O}}_{X,p} \simeq k[[x_1, \cdots, x_{d+1}]]/(Q(x_1, \cdots, x_{d+1}))$$

where Q is nondegenerate quadric.

Theorem 9.4. *If* $\mathrm{char}(k) = 0$*, then $(X, \mathcal{L})$ admits a Lefschetz pencil. If* $\mathrm{char}(k) > 0$*, then $(X, \mathcal{L}^{\otimes m})$ admits a Lefschetz pencil for a suitable choice of m.*

Proof. (Sketched). Let $X \neq \mathbb{P}^N$ is connected, $\dim X = d$. Consider the incidence correspondecne:

$$\Phi \coloneqq \{(x, H) : x \in H \cap X, T_x X \subset H\}$$

We thus have two projections:

$$p_1 : \Phi \to X, \quad p_2 : \Phi \to X^\vee \subset \mathbb{P}^N$$

We know $p_1 : \Phi \to X$ is a projective bundle with fibres $\mathbb{P}^{N-d-1}$. Hence Φ is irreducible, complete and $\dim = N - 1$.

p_2 has fibres $p_2^{-1}(H) = \mathrm{Sing}(H \cap X)$ □

Definition 9.5. The Lefschetz fibration associated to $(X, \mathcal{L}, D)$ is the incidence correspondence $X^* = \mathrm{Bl}_{A\cap X} X \xrightarrow{\pi} D \simeq \mathbb{P}^1$

Lemma 9.6. *$\pi : X^* \to D$ is proper, flat and has a section.*

9.3 Symplectic Monodromy

Let $X_0 \subset \mathbb{P}^N$ be a nice variety of dimensional $n+1$ over $\mathbb{F}_q$.

Let $S \subset \mathbb{P}^1$ be the "bad" points. Set $U_0 = \mathbb{P}^1 \backslash S_0$. The map $\pi : X^* \to \mathbb{P}^1$ is smooth and proper over U, hence by base change

$$\mathcal{V} \coloneqq R^n \pi_* \mathbb{Q}_\ell$$

is an ℓ-adic local system over U. This equivalent to the data of

$$\pi_1^{\text{ét}}(U) \to \mathrm{GL}(\mathcal{V}_{\overline{u}})$$

THe mildnessof singularity of $X^* \to \mathbb{P}^1$ will implythe representation is tame.

Definition 9.7. Let $\mathbb{D} \coloneqq \mathrm{Spec}\ (\hat{\mathcal{O}}_{\mathbb{P}^1,s}) \to \mathbb{P}^1$, $\mathbb{D}^0 \coloneqq \mathrm{Spec}\ (\mathrm{Frac}(\hat{\mathcal{O}}_{\mathbb{P}^1,s})) \to U$. The inertia I_S is the image of $\pi_1(\mathbb{D}^0) \hookrightarrow \pi_1(U)$.

Let $j : U \to \mathbb{P}^1$, and $\mathcal{F} = R^n \pi_* \mathbb{Q}_\ell$. $\mathcal{F} \to j_* j^* \mathcal{F}$ is an iso iff for all $s \in S$, the specialization map $\mathcal{F}_s \to \mathcal{F}_{\overline{u}}$ is injective and has image $\mathcal{F}_{\overline{u}}^{I_S}$.

The space of vanishing cycle at $s \in S$ is

$$E_s \coloneqq \ker(H^n(X_{\overline{u}}, \mathbb{Q}_\ell(n))^\vee \xrightarrow{\text{cospecialization}(n)^\vee} H^n(X_s, \mathbb{Q}_\ell(n))^\vee)$$

The upshot is E_s is one-dimensional, with generator called $\delta_s(-m)$.

Theorem 9.8 (Picard-Lefschetz)**.** *Let $\sigma_s \in I_s$. Then for all $x \in H^n(X_{\overline{u}}, \mathbb{Q}_\ell)$ have*

$$\sigma_s(x) = x \pm t(\sigma_s)(x \cup \delta_s)\delta_s$$

10 Deligne's Proof Part III

Recall the statement:

If $X_0/\mathbb{F}_q$ is smooth and proper, then the Frobenius F acting on $H^r(X, \mathbb{Q}_\ell)$ has eigenvalues whose complex conjugates α all have $|\alpha| = q^{\frac{r}{2}}$

We've seen it's enough to show: If $\dim X_0 = 2m + 2 = n + 1$, then the Frobenius action on the middle cohomology $H^{n+1}(X, \mathbb{Q}_\ell)$ is rational and its eigenvalues satisfy:

$$q^{\frac{n}{2}} < |\alpha| < q^{\frac{n}{2}+1}$$

Denote $(*)(n)$ the statement that $F : V \to V$ has eigenvalues α satisfying $q^{\frac{n}{2}} < |\alpha| < q^{\frac{n}{2}+1}$

Lemma 10.1. *Let $F : V \to V$.*

1. *If V satisfies $(*)(n)$ and W stable under F, then W and V/W both satisfy $(*)(n)$*
2. *If there is a filtration $V \supset V_1 \cdot \supset V_r$ stable under F, and each subquotient satisfies $(*)(n)$, then so does V.*

Let's fix some notation for Lefschetz pencil.

- $D \simeq \mathbb{P}^1$ is a Lefschetz pencil for $X \subset \mathbb{P}^N$
- $0, \infty \in \mathbb{P}^1$, its axis is $A = H_0 \cap H_\infty$.
- We define $X^* = \mathrm{Bl}_A X$ blowing up of X in $A \cap X$.
- The diagram

$$\begin{array}{ccc} X & \xleftarrow{\varphi} & X^* \\ & & \downarrow{\scriptstyle \pi} \\ & & D = \mathbb{P}^1 \end{array}$$

  such that the fibre of $t \in D = \mathbb{P}^1$ is $X_t = X \cap H_t$

The idea is to show RH for the nicer space X^*.

Lemma 10.2. *It suffices to show $(*)$ for X^*.*

Proof. Set $N_{X/A\cap X}$ be the normal bundle of $A \cap X$ in X. Then:

$$\varphi^{-1}(A \cap X) \simeq \mathbb{P}(N_{X/A\cap X})$$

Also $N \cap X$ has codim 2 in X, so $N_{X/A\cap X}$ has rank 2. Thus by using Chern classes we get

$$H^*(\varphi^{-1}(A \cap X), \mathbb{Q}_\ell) \simeq H^*(A \cap X, \mathbb{Q}_\ell) \oplus H^{*-2}(A \cap X, \mathbb{Q}_\ell)(-1)$$

(this is a general property of projective bundle). By proper base change, we have:

1. The map $\mathbb{Q}_\ell \to \varphi_* \mathbb{Q}_\ell$ is an isomorphism.
2. $R^2\varphi_* \mathbb{Q}_\ell$ has support on $A \cap X$.
3. $R^r \varphi_* \mathbb{Q}_\ell = 0$ for $r \neq 0, 2$.

Thus the Leray spectral sequence

$$H^p(X, R^q \varphi_* \mathbb{Q}_\ell) \Rightarrow H^{p+q}(X^*, \mathbb{Q}_\ell)$$

degenerates. Hence we see:

$$H^*(X^*, \mathbb{Q}_\ell) = H^*(X, \mathbb{Q}_\ell) \oplus H^{*-2}(A \cap X, \mathbb{Q}_\ell)(-1)$$

We conclude by applying previous lemma. □

Now look at Leray spectral sequence for $\pi : X^* \to \mathbb{P}^1$.

$$E_2^{p,q} = H^p(\mathbb{P}^1, R^q\pi_*\mathbb{Q}_\ell) \Rightarrow H^{p+q}(X^*, \mathbb{Q}_\ell)$$

The terms contributing to H^{n+1} are:

$$H^0(\mathbb{P}^1, R^{n+1}\pi_*\mathbb{Q}_\ell), H^1(\mathbb{P}^1, R^n\pi_*), H^2(\mathbb{P}^1, R^{n-1}*)$$

Only need to show $(*)(n+1)$ for these groups.

Let's start by computing $H^2(\mathbb{P}^1, R^{n-1}\pi_*\mathbb{Q}_\ell)$. One can show the sheaf $R^{n-1}\pi_*\mathbb{Q}_\ell$ is constant on $\mathbb{P}^1$. Thus:

$$H^2(\mathbb{P}^1, R^{n-1}\pi_*\mathbb{Q}_\ell) \simeq (R^{n-1}\pi_*\mathbb{Q}_\ell)_t(-1) \simeq H^{n-1}(X_t, \mathbb{Q}_\ell)(-1)$$

Let $Y_0 \subset X_{t,0}$ be a smooth hyperplane section. So $X_t \backslash Y$ is affine. Using the excision sequence we see

$$H^{n-1}(X_t, \mathbb{Q}_\ell) \hookrightarrow H^{n-1}(Y, \mathbb{Q}_\ell)$$

By induction we see $(*)$ holds for Y, thus $(*)(n+1)$ holds for $H^2(\mathbb{P}^1, R^{n-1}\pi_*\mathbb{Q}_\ell)$

The treatment for $H^0(\mathbb{P}^1, R^{n+1}\pi_*\mathbb{Q}_\ell)$ is similar. It remains to show $(*)(n+1)$ for H^1.

Let $S \subset D = \mathbb{P}^1$ be the locus of singular fibres. Let $U = \mathbb{P}^1 - S$ and $u \in U$ be a geometric point. Let $V = (R^n\pi_*\mathbb{Q}_\ell)_u$. Let $E \subset V$ be generated by vanishing cycles. There exists a symplectic pairing

$$\psi : V \times V \to \mathbb{Q}_\ell(-n)$$

by cup product. We denote $E^\perp$ be the orthogonal complement of E under ψ.

Remark. *If we have Hard Lefschetz for (ℓ-adic cohomology), then can show $E \cap E^\perp = 0$.*

Let's consider the filtration:

$$V \supset E \supset E \cap E^\perp$$

The Picard-Lefschetz formula will imply the action of $\pi_1^{\text{tame}}(U)$ on V/E and $E \cap E^\perp$ are trivial. Hence we get filtration of sheaves on U

$$\mathcal{V} \supset \mathcal{E} \supset \mathcal{E} \cap \mathcal{E}^\perp$$

and $\mathcal{V}/\mathcal{E}$ and $\mathcal{E} \cap \mathcal{E}^\perp$ are cosntant. Applying j_* for $j : U \hookrightarrow \mathbb{P}^1$ gives:

$$R^n\pi_*\mathbb{Q}_\ell \supset j_*\mathcal{E} \supset j_*(\mathcal{E} \cap \mathcal{E}^\perp)$$

Lemma 10.3 (Rationality Lemma)**.** *For all $x \in U$, the Frobenius action F_x on $\mathcal{E}/\mathcal{E} \cap \mathcal{E}^\perp$ is rational.*

Once we know this, by Kazhdan-Margulis Theorem, we know the image of $\pi_1^{\text{tame}}(U)$ is open in $\text{Sp}(E/E \cap E^\perp, \psi)$.

Then we are in a good shape to apply the Main lemma. Hence we know the Frobenius action on $H^1(\mathbb{P}^1, j_*(\mathcal{E}/\mathcal{E} \cap \mathcal{E}^\perp))$ satisfies $(*)(n+1)$.

11 Weights and Statement of Weil II

Question: Given $f : X \to Y$ of schemes over $\mathbb{F}_q$, can we relate Frobenius eigenvalues of $\mathcal{F}$ on $\overline{X}$, with Frobenius eigenvalues on $R^i f_* \mathcal{F}$?

11.1 Weights

Definition 11.1. Let $X/\mathbb{F}_q$ be a scheme of finite type, and $\mathcal{F}$ a $\overline{\mathbb{Q}_\ell}$ a sheaf on X, given $\tau : \overline{\mathbb{Q}_\ell} \to \mathbb{C}$.

1. $\mathcal{F}$ is τ-pure of weight w if for all $x \in X(\overline{\mathbb{F}_q})$ the eigenvalues α of $\mathcal{F}_x^{\deg x}$ acting on $\mathcal{F}_x$ satisfy: $\tau(\alpha) = (q^{\deg x})^{w/2}$.
2. We say $\mathcal{F}$ is τ-mixed, if there exists a filtration such that each subquotient is τ-pure.
3. We say $\mathcal{F}$ is pure if it's τ-pure for all τ
4. $\mathcal{F}$ if there exists a filtration with subquotients pure.
5. $\mathcal{F}$ is τ-mixed of weight $\leq w$ if every $\mathcal{F}_i/\mathcal{F}_{i-1}$ is τ-pure of weight $\leq w$.

Example. *The Tate twist $\mathbb{Q}_\ell(1)$ is pure of weight -2. $\overline{\mathbb{Q}_\ell}$ is pure of weight 0.*

Theorem 11.2 (Deligne)**.** *Let $f : X \to Y$ be a morphism of schemes of finite type over $\mathbb{F}_q$. If $\mathcal{F}$ is a constructible $\overline{\mathbb{Q}_\ell}$ sheaf of X mixed of weight $\leq n$. Then, for all $i \geq 0$, $R^i f_! \mathcal{F}$ mixed of weight $\leq n + i$*

Remark. $f_! \mathcal{F}(U) = \{s \in \mathcal{F}(f^{-1}U) \mid \text{supp}(s) \text{ proper}/U\}$

Corollary 11.3. *When $Y = \text{Spec } \mathbb{F}_q$. Then $R^i f_! \mathcal{F} = H_c^i(\overline{X}, \mathcal{F})$ mixed of weight $\leq n + i$. If moreover X is smooth, and $\mathcal{F}$ is mixed of weight $\geq m$, then $H^i(\overline{X}, \mathcal{F})$ mixed of weight $\geq m + i$.*

Corollary 11.4. *If X smooth, $\mathcal{F}$ pure of weight n. Then*

$$\text{im } (H_c^i(\overline{X}, \mathcal{F}) \to H^i(\overline{X}, \mathcal{F}))$$

is pure of weight $n + i$. In particular, if X is smooth and proper, then $H^i(X, \mathcal{F})$ pure of weight $n + i$.

Remark. *For $\mathcal{F} = \overline{\mathbb{Q}_\ell}$, this is Weil I.*

11.2 Basic Functoriality of Pure Sheaves

1. Suppose $f : X \to Y$, $\mathcal{F}$ is a sheaf on Y. Assume $\mathcal{F}$ is pure of weight n, then $f^* \mathcal{F}$ pure of weight n. This is because purity is a condition on stalk.
2. For $f : X \to Y$ finite, $\mathcal{F}$ sheaf on X. If $\mathcal{F}$ pure of weight n, so does $f_* \mathcal{F}$. Similar as above.

3. Let $X/\mathbb{F}_q$, $\mathbb{F}_{q'}/\mathbb{F}_q$ finite. Let $f : X_{\mathbb{F}_{q'}} \to X$ be the base change map. If $\mathcal{F}$ on X pure of weight $n \iff f^*\mathcal{F}$ pure of weight n.

4. $\mathcal{F}$ pure (resp. mixed), so is any subquotients.

5. Suppose
$$0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$$
If $\mathcal{F}, \mathcal{H}$ mixed, then so is $\mathcal{G}$.

6. The previous two facts combined tell me if $\mathcal{F} \to \mathcal{G} \to \mathcal{H}$ exact, $\mathcal{F}, \mathcal{H}$ mixed, then so does $\mathcal{G}$.

For Weil II, we have the following setup. Let $f : X \to Y$ finite type over $\mathbb{F}_q$, with τ-mixed sheaf $\mathcal{F}$ with weights $\leq n$. We know:

1. make arbitrary field extensions

2. The theorem is trivial for f quasi-finite

3. Given $0 \to \mathcal{F} \to \mathcal{G} \to \mathcal{H} \to 0$, with $\mathcal{F}, \mathcal{H}$ satisfies Weil II, then so does $\mathcal{G}$. We can see this by looking at the LES:
$$R^i f_! \mathcal{F} \to R^i f_! \mathcal{G} \to R^i f_! \mathcal{H}$$

4. If we have $U \hookrightarrow X \hookleftarrow Z$, and $\mathcal{F}|_U$, $\mathcal{F}|_Z$ satisfies Weil II, then $\mathcal{F}$ satisfies Weil II. We see this by looking at the distinguished triangle:
$$0 \to j_! j^* \mathcal{F} \to \mathcal{F} \to i_! i^* \mathcal{F} \to 0$$

5. If we know $X \xrightarrow{f} Y \xrightarrow{g} Z$, with f, g satisfy Weil II for f, g, then so does $g \circ f$. Look at spectral sequence:
$$E_2^{p,q} = R^p g_! (R^q f_! \mathcal{F}) \Rightarrow R^{p+q} (gf)_! \mathcal{F}$$

6. Can replace X, Y by $X_{\text{red}}, Y_{\text{red}}$.

7. Use Noetherian induction we can assume $f : X \to Y$ relative dimension ≤ 1. Eventually, I can assume $f : X \to Y$ smooth, with fibres nonempty, geometrically connected irreducible smooth curves.

11.3 τ-real Sheaves

Definition 11.5. A sheaf $\mathcal{F}$ is τ-real if for all $x \in |X|$, the action of Frobenius on $\mathcal{F}_x$ has characteristic polynomial $\tau(\det(1 - \text{Frob } t)) \in \mathbb{R}[t] \subset \mathbb{C}[t]$.

Lemma 11.6. *Let $\mathcal{G}$ be a local system of schemes over $\mathbb{F}_q$, τ-pure of weight w. Then $\mathcal{G}$ is a direct summand of τ-real, τ-pure local system $\mathcal{F}$ of weight w.*

Proof. Choose $b \in \overline{\mathbb{Q}_\ell}$ such that $|\tau(b)| = q^w$. I set $\mathcal{F} = (\mathcal{G}^\vee \otimes \mathcal{L}_b) \oplus \mathcal{G}$. This is real. □

We'll see for $\mathcal{F}$ local system on X, then τ-real implies τ-mixed.

12 Local Monodromy

12.1 Grothendieck's Monodromy Theorem

Let $X/\mathbb{Q}$ be proper smooth variety.

Let R be Henselian DVR, $K = \mathrm{Frac}(R)$, $k = R/\mathfrak{m}$ with $\mathrm{char}(k) = p$. Fix $\overline{K} \supset K$ algebraic closure, which induces $\overline{k} \supset k$. We have the monodromy exact sequence:

$$1 \to I_K \to \mathrm{Gal}(\overline{K}/k) \to \mathrm{Gal}(\overline{k}/k) \to 1$$

I also have an exact sequence for inertia:

$$1 \to P_K \to I_K \to \prod_{\ell \neq p} \mathbb{Z}_{\ell(1)} \to 1$$

P_K is called the wild inertia.

Let me set:

$$t_\ell : \prod_{\ell' \neq p} \mathbb{Z}_{\ell'}(1) \to \mathbb{Z}_\ell(1)$$

so $\ker t_\ell = \prod_{\ell' \neq p,\ell} \mathbb{Z}_{\ell'}(1)$, this fits into:

$$1 \to P_K \to P_{K,\ell} \to \prod_{\ell' \neq p,\ell} \mathbb{Z}_{\ell'}(1) \to 1$$

We've seen lisse $\mathbb{Q}_\ell$ sheaves $\mathcal{F}_0$ corresponds to continuous representations of $\pi_1^{\text{ét}}(X_{\overline{\mathbb{F}}_q})$.

Theorem 12.1 (Grothendieck)**.** *Let $\rho \in \mathrm{Rep}_{\mathbb{Q}_\ell}(G_K)$. Assume no finite extension of k contains every ℓ^n-th root of unity. Then there exists an open subgroup $J \subset I_K$ such that $\rho(\sigma)$ is unipotent for all $\sigma \in J$. This is saying ρ is potentially semi-stable*

Lemma 12.2. *Let G be a profinite group, pro-order prime to ℓ. Then every $\rho : G \to \mathrm{GL}_n(\mathbb{Q}_\ell)$ has finite image.*

Lemma 12.3. *There exists unique $N : V(1) \to V$ nilpotent such that*

$$\rho(\sigma) = \exp(N \circ t_\ell(\sigma))$$

for all $\sigma \in J \subset I_K$. This is called **monodromy operator**.

Lemma 12.4. *Given $N : V \to V$ nilpotent endomorphism. There exists a unique increasing filtration $M_\bullet$ such that:*

1. *$N(M_iV) \subset M_{i-2}V$*
2. *N^r induces an isomorphims $\mathrm{gr}_r(V) \simeq \mathrm{gr}_{-r}(V)$.*

12.2 Weight-Monodromy Conjecture

Theorem 12.5 (Main)**.** *Let $f : X \to Y$ be a finite type scheme over $\mathbb{F}_q$. Let $\mathcal{F}$ be a constructible $\mathbb{Q}_\ell$ sheaf, mixed of weight $\leq n$. Then for all $i \geq 0$, then the sheaf $R^i f_! \mathcal{F}$ is mixed of weight $\leq n + i$.*

A first reduction is that we may assume $X \to Y$ is a smooth projective family of curves.

Theorem 12.6 (Main case)**.** *$Y_0/\mathbb{F}_q$ smooth geometrically irreducible curve. Let $j_0 : U_0 \hookrightarrow Y_0$ open dense subscheme complement of S_0. Let $\mathcal{F}_0$ be a lisse $\overline{\mathbb{Q}}_\ell$-sheaf, pointwise ι-pure of weight n on U_0. Then Then $\mathrm{gr}_i^{M_\bullet} \mathcal{F}_{0,\overline{\eta}}$ (acted by $\mathrm{Gal}(\overline{\eta}/\eta)$) is ι-pure of weight $n + i$.*

Let's introduce Weight filtration. Let $k = \mathbb{F}_q$, we have:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & I_K & \longrightarrow & W_K & \longrightarrow & W_k & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & I_K & \longrightarrow & G_K & \longrightarrow & G_k \simeq \hat{\mathbb{Z}} & \longrightarrow & 1 \end{array}$$

Given $N : V(1) \to V$ monodromy operator for all (ρ, V).

Definition 12.7 (Weight Filtration)**.** Suppose for $\iota : \overline{\mathbb{Q}}_\ell \simeq \mathbb{C}$, the ι-weights are integers. Then there exists unique increasing filtration $W_\bullet$ of V, stable under W_K, such that $\mathrm{gr}_i^{W_\bullet}(V)$ is ι-pure of weight i. This is the weight filtration.